

A

Major Project Report on

DEVELOPMENT OF RIPPED FRUIT PLUCKING ROBOT

Submitted for partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

MECHANICAL ENGINEERING

By

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20K81A0310

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Certificate

This is to certify that the project entitled **“DEVELOPEMENT OF RIPPED FRUIT PLUCKING ROBOT”** is being submitted by Manohar Eedarada (20K81A0310) in fulfilment of the requirement for the award of degree of **BACHELOR OF TECHNOLOGY IN MECHANICAL ENGINEERING** is recorded of bonafide work carried out by them. The results embodied in this report have been verified and found satisfactory.

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DECLARATION

We, the students of **“Bachelor of Technology in Department of Mechanical Engineering”**, session: 2020-2024, **St.Martin’s Engineering College**, Dhulapally, Kompally, Secunderabad, hereby declare that the work presented in this project work entitled **“DEVELOPEMENT OF RIPPED FRUIT PLUCKING ROBOT”** is the outcome of our own bonafide work and is correct to the best of our knowledge and this work has been undertaken taking care of Engineering Ethics. The result embodied in this project report has not been submitted to any university for award of any degree.

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Acknowledgement

The satisfaction and euphoria that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose encouragement and guidance have crowded our efforts with success.

First and foremost, we would like to express our deep sense of gratitude and indebtedness to our College Management for their kind support and permission to use the facilities available in the Institute.

We especially would like to express our deep sense of gratitude and indebtedness to **Dr. P. SANTOSH KUMAR PATRA**, Group Director, St. Martin's Engineering College Dhulapally, for permitting us to undertake this project.

We wish to record our profound gratitude to **Dr. M. SREENIVAS RAO**, Principal, St. Martin's Engineering College, for his motivation and encouragement.

We are also thankful to **Dr. D.V. SREEKANTH**, Head of the Department, Mechanical Engineering, St. Martin's Engineering College, Dhulapally, Secunderabad, for his support and guidance throughout our project as well as Project Coordinator **K. HEMALATHA**, Assistant Professor, Mechanical Engineering, department for his valuable support.

We would like to express our sincere gratitude and indebtedness to our project supervisor **Dr. A. UDAY KUMAR**, Assistant Professor, Mechanical Engineering, St. Martins Engineering College, Dhulapally, for his support and guidance throughout our project.

Finally, we express thanks to all those who have helped us successfully complete this project. Furthermore, we would like to thank our family and friends for their moral support and encouragement. We express thanks to all those who have helped us in successfully completing the project.

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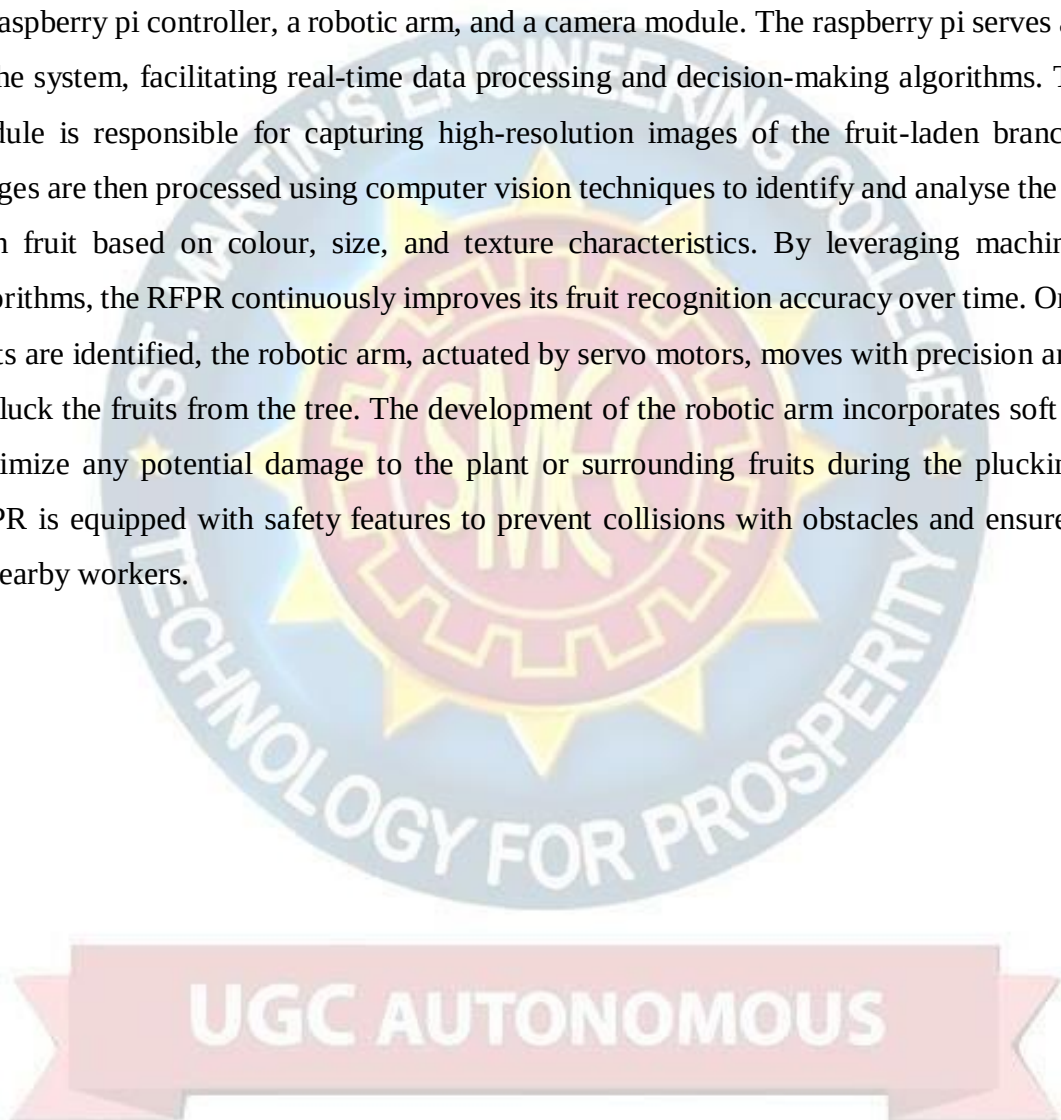
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ABSTRACT

Ripped Fruit Plucking Robot designed using the raspberry pi controller platform. The RFPR is a sophisticated robotic system that combines computer vision and precise manipulation to autonomously detect and harvest ripe fruits from trees. The core components of the RFPR include an raspberry pi controller, a robotic arm, and a camera module. The raspberry pi serves as the brain of the system, facilitating real-time data processing and decision-making algorithms. The camera module is responsible for capturing high-resolution images of the fruit-laden branches. These images are then processed using computer vision techniques to identify and analyse the ripeness of each fruit based on colour, size, and texture characteristics. By leveraging machine learning algorithms, the RFPR continuously improves its fruit recognition accuracy over time. Once the ripe fruits are identified, the robotic arm, actuated by servo motors, moves with precision and delicacy to pluck the fruits from the tree. The development of the robotic arm incorporates soft grippers to minimize any potential damage to the plant or surrounding fruits during the plucking process. RFPR is equipped with safety features to prevent collisions with obstacles and ensure the safety of nearby workers.



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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Fruit growers are facing an increasingly severe labor shortage due to the labor work force's diminishing interest in agriculture. The problem is exacerbated by recent international travel restrictions in pandemic conditions, which have dramatically limited agricultural productivity due to the unavailability of migrant workers. As a result, tons of fresh produce were left unharvested and rotting on fields where farms had long relied on seasonal overseas workers. Harvesting robots can counter labor shortages and provide an economically viable solution to rising labor costs. Robots can also be a tool for applying precision agriculture. Specifically, the application of the harvesting robots makes the harvesting process controllable, traceable, and customizable, meanwhile harvesting robots can gather, process and analysis temporal, spatial, and individual data of the target fruits as the environment information and combines it with other information to provide support to decision-makers. The technology of robotic harvesting has been actively developed over the past three decades. In terms of the technology status of the harvesting robot, the robotic systems analyzed can be categorized into two groups: fully integrated systems, and subsystems used in harvestings, such as vision, gripper, and control. Since each of these subsystems is quite a broad research area by itself, this paper will be focused on reviewing integrated systems.



Fig 1.1 Fruit Plucking Robot

examined mechanical harvesting systems as well as robotic harvesting applications before 2011 and discussed the machine vision approaches used in harvesting robots.

Characterized the crop environment, summarized quantitative performance indicators for crop harvesting and conducted a thorough comparison on existing robots before 2014. In the last 6 years, Bachchan emphasized analyzing the design strategies behind various applications, Zhao et al. focused on vision control techniques and their potential applications in fruit or vegetable harvesting robots, Shamshir et al. and Fountas et al. both collected and presented around twenty recent robotic harvesting applications as one section of their review paper on agricultural robots, and Jia et al. and Zhang reviewed apple harvesting robots and harvest assist platforms. There are two major gaps in this work attempts to fill, regarding a state-of-the-art review of harvesting robots. Firstly, to the best of the authors' knowledge, none of the existing reviews in the past six years have compared the robot performance in recent applications to three key performance indicators that are critical for potential commercial viability: harvest success rate, cycle time, and damage rate. Only one review paper has covered performance indicators among 50 applications, but there was a lack of performance data to provide a conclusive comparison between harvesting robots at the time. Secondly, there is a lack of systematic investigation on the underlying reasons for the inadequate performance of existing harvesting robots, relative to progress towards commercialization. This paper will summarize all the challenges reported in recent literature and analyses the origin of these challenges from the point of view of both environment and requirements of fruit growers. With this investigation, a connected map of the challenges and corresponding research topics that link the environmental challenges of harvesting with customer requirements is identified for the first time. The objective of this work is to present a systematic review of the status of the fruit harvesting technology and to propose a standardized framework to quantitatively compare the state-of-the-art robots in fruit harvesting and to gauge their commercial potential. The framework of this research involves the following steps I) the review protocol regarding the inclusion and exclusion criteria; exploration and selection of related data; classification of selected data; analysis and interpretation of the selected data; and reporting findings and results. To investigate the similarities and the differences of fruit harvesting robots among projects and fruit varieties, the data in the literature were selected and analyzed against the following research philosophy: The included data/studies are collected from multidisciplinary databases and open online sources.



Fig: 1.2 Ripped Fruit Plucking Robot

This review work only included data and references related to fruit farming, applications that are closely related to integrated fruit harvesting robotic systems that underwent field tests and operations within the last 20 years. Fruit harvesting applications published before the year 2000 are excluded. To make the technology; These data comparable, those integrated prototypes that have not been tested in the field are included in the discussion part but excluded in the system analysis section; Studies focusing on only subsystems of the selected harvesting robots are also included in this work to present the audience a systematic review of the state of the art robotic harvesting technology; The included harvesting robots were then reviewed against the aspects of the fruit detection methods, fruit detachments methods and obstacle handling approaches, the performance of the included robots were then quantitatively compared on three most used indicators: harvest success rate, cycle time/overall speed and damage rate. The challenges were extracted directly from the literature involved in this work to present the robot developers' original records of the issues encountered, and the discussions

1.2 Background

Two types of harvesting approaches have been implemented by agricultural practitioners to reduce orchard overheads on labor expenses: selective harvesting and bulk harvesting. Selective harvesting is a harvesting method for robotic systems utilizing robotic manipulators with end-effectors for grasping. These are typically mounted on a mobile platform with an end-effector and machine vision to selectively pick individual ripe fruit. Since robotic systems may potentially combine the efficiency of machines with the selectivity of humans on a long-term basis, it is believed that the robotic harvesting approach has the potential to completely

replace human pickers. Therefore, it has attracted extensive attention from both academia and industry and has emerged as the preferred method of robotic harvesting amongst fruit growers. The ongoing rapid development of artificial intelligence and robotics technologies is paving the last mile of commercial robotic harvesting; hence the focus of this review will be on robotic systems utilizing selective harvesting. The other method of harvesting is called the bulk harvesting method and is based on the principle of applying vibration to the fruit tree to force separate the fruit. It has been adopted by growers of various fruits such as apples, oranges, and cherries. Although mass harvesting systems come with high efficiency, there are significant disadvantages. Growers have raised concerns over the excessive damage caused by the machines to both canopies and the fruit. Since fruit damage affects its market acceptance, the research on reducing the bulk harvesting damage remains an active field. Another disadvantage of bulk harvesting is that the quality of picked fruits can vary considerably, where less mature fruits are inevitably harvested together with over mature fruits. Co-ordinating rates of maturity of fruit across an entire orchard is not a trivial task, and under a bulk harvesting scheme, the timing of harvest.

1.3 Working principles

A fruit plucking robot is a highly sophisticated agricultural automation system that revolutionizes the age-old practice of fruit harvesting. These machines are engineered with a combination of cutting-edge technologies and principles to harvest ripe fruits efficiently and systematically from orchards. Central to their operation is their sensory and detection systems, which often include advanced vision systems, such as high-resolution cameras and computer vision algorithms. These components enable the robot to identify and locate ripe fruits by analyzing their color, shape, and size, ensuring that only the mature and ready-to-harvest fruits are picked. Furthermore, many fruit plucking robots incorporate laser or Lidar sensors to generate precise 3D maps of the orchard. This mapping capability is essential for both navigation and avoiding obstacles, as it provides the robot with a real-time understanding of the orchard's layout. The machine's mobility is a critical aspect of its operation, and it is typically equipped with wheels or tracks for efficient movement. Some designs even employ autonomous vehicles with multiple wheels or tracks to enhance stability and maneuverability, ensuring the robot can navigate the often uneven and challenging terrain of orchards. The heart of the fruit plucking robot lies in its plucking mechanism. Most robots use soft, flexible grippers designed to grasp the fruit delicately and securely without causing any damage. These grippers are often adaptable to

various fruit shapes and sizes, providing a versatile solution for different types of fruit-bearing trees. In some cases, the robots employ vibration or shaking mechanisms to gently dislodge fruits from their branches, while others may utilize pruning tools or shears to cut fruit stems when plucking. These plucking mechanisms aim to preserve the quality and integrity of the harvested fruits. Control and automation are at the core of the fruit plucking robot's operation. Advanced AI and machine learning algorithms are employed to make real-time decisions about fruit ripeness, navigation through the orchard, and the precise moment to initiate plucking. Communication with a central control system or remote operators allows for monitoring and adjustments to ensure optimal performance. These robots leverage state-of-the-art technologies to operate autonomously, enhance efficiency, and maintain fruit quality, providing substantial benefits to farmers while addressing labor shortages and increasing the overall yield and quality of harvested fruits.



Fig 1.3 Plucking Robot

Fruit plucking robots are a marvel of modern agricultural technology, integrating a multitude of advanced principles and techniques to streamline the fruit harvesting process. At their core, these robots rely on a combination of sensors, such as vision systems, GPS, and IMU, to identify and locate ripe fruits in orchards, considering factors like color, shape, and size. Their mobility is ensured by wheels, tracks, or specialized locomotion systems, adapting to various terrains and orchard layouts. The plucking mechanisms are diverse, featuring soft and adaptable grippers for gentle fruit extraction, as well as vibration or shaking technologies to safely dislodge ripe fruits. Data collection, including quality assessment and yield estimation, is facilitated through an array of sensors, contributing to efficient orchard management. Energy-efficient designs and sustainability considerations are embedded in these robots, making use of advanced battery technology and, in some cases, renewable energy sources like solar panels. As research and development continue to advance, fruit plucking robots are expected to become even more

autonomous, accurate in ripeness detection, and adaptable to different fruit varieties and orchard conditions, heralding a brighter, more efficient future for fruit harvesting in agriculture.

Fruit plucking robots are intricate agricultural marvels that embody a fusion of advanced technologies and principles to streamline the fruit harvesting process. These robots, equipped with sophisticated sensors and vision systems, make real-time decisions based on data analytics, determining fruit ripeness and precise locations in orchards. Their mobility systems, whether wheels, tracks, or specialized locomotion methods, offer adaptability to varied terrains and orchard layouts. Furthermore, these robots employ user-friendly interfaces and remote-control capabilities, bridging the gap between automation and human oversight. Machine learning and AI empower them to continuously improve their fruit recognition and handling precision over time. Integration with broader farm management systems and obstacle avoidance technology ensures seamless navigation and comprehensive orchard management. Their adaptability to different fruit types, post-harvest handling capabilities, wireless communication, and scalability in larger orchards enhance their versatility. These robots have economic and labor-saving benefits, optimizing fruit harvesting schedules, reducing waste, and mitigating labor shortages, making them a significant contributor to modern agriculture's efficiency and sustainability. As research and development progress, fruit plucking robots are poised to redefine the future of fruit cultivation. Fruit plucking robots represent a remarkable convergence of cutting-edge technology and agricultural necessity. These robotic systems have evolved to integrate an array of intricate working principles and technologies, each contributing to their effectiveness in automating fruit harvesting processes. The foundation of their operation lies in their sensory and detection systems, encompassing vision systems equipped with high-resolution cameras and advanced computer vision algorithms. This enables the robots to discern the optimal ripeness of fruits by analyzing various factors, including color, shape, and size, ensuring that only the ripest produce is selected for harvesting. Their mobility is equally complex and adaptable, with some robots equipped with wheels, tracks, or even more specialized locomotion systems designed to tackle challenging orchard terrains and layouts.

The plucking mechanisms within these robots are designed to execute their tasks with precision and care. They often incorporate soft and compliant grippers that mimic the delicate touch of a human hand, ensuring that fruits are harvested without any damage or bruising. However, these robots aren't limited to a one-size-fits-all approach, as they may use various methods such as vibration or shaking technologies to gently dislodge ripe fruits, or they may employ pruning tools for cutting fruit stems during the plucking process. These mechanisms are versatile and adaptable

to suit different fruit types, shapes, and sizes, contributing to the overall efficiency and quality of fruit harvesting. Fruit plucking robots also rely heavily on real-time decision-making capabilities, facilitated by advanced algorithms that continuously analyze the data from their sensors. This real-time adaptability ensures that the robot can respond to changing conditions and adapt its movements to navigate through orchards with precision, all while preserving the fruit quality. These robots often come equipped with user-friendly interfaces that allow for remote control and monitoring, which is vital for addressing unforeseen challenges or for overseeing the operation as needed. Machine learning and artificial intelligence play a significant role in enhancing the efficiency of these robots, enabling them to learn from historical data and make more accurate decisions over time, from identifying fruit characteristics to optimizing harvesting schedules. Moreover, fruit plucking robots are designed with a comprehensive approach to data collection, encompassing quality assessment and yield estimation through an array of sensors. This data, used for real-time decision-making and reporting, ensures that the ripest and highest-quality fruits are harvested. The energy efficiency and sustainability of these robots are also carefully considered, with designs focused on efficient energy consumption and the use of advanced battery technology or renewable energy sources like solar panels for power. Furthermore, these robots are equipped with obstacle avoidance technology that employs sensors such as ultrasonic sensors and LiDAR to detect and circumvent potential obstacles within the orchard, ensuring safe and efficient navigation. Their adaptability to different fruit types is a testament to their versatility, and some robots can even offer post-harvest handling capabilities, including sorting and packaging of harvested fruits. The future of fruit plucking robots continues to evolve, with ongoing research and development aimed at further enhancing their autonomy, accuracy in ripeness detection, adaptability to different fruit varieties and orchard conditions, and overall efficiency in fruit harvesting. In essence, fruit plucking robots have not only revolutionized the agricultural landscape but have positioned themselves as a promising solution to labor shortages, waste reduction, and the optimization of fruit harvesting practices, contributing significantly to modern agriculture's efficiency and sustainability.

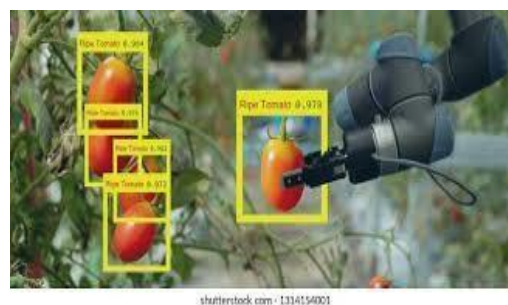


Fig 1.4 Recognition Cam View

1.4 Applications

Ripe fruit plucking robots find applications in various sectors, primarily in agriculture and related industries. These robots address labor shortages, improve fruit harvesting efficiency, and enhance the overall quality of the produce. Here are some key applications of ripe fruit plucking robots:

1. **Orchards and Fruit Farms:** The primary application of fruit plucking robots is in commercial orchards and fruit farms. These robots are used to harvest a wide range of fruits, such as apples, pears, peaches, and citrus fruits, where labor-intensive harvesting is traditionally required. They increase efficiency, reduce labor costs, and improve yield by ensuring that ripe fruits are harvested at the right time.
2. **Vineyards:** Fruit plucking robots have found applications in vineyards for harvesting grapes. Grapes need to be harvested at the precise moment of ripeness to produce high-quality wine, and robots can ensure that this timing is exact. They are also less likely to damage the vines compared to manual harvesting.
3. **Greenhouses:** In controlled environments like greenhouses, robots can be used to pick ripe fruits such as tomatoes, cucumbers, and bell peppers. Their ability to operate in confined spaces and their non-disruptive harvesting techniques make them particularly useful in such settings.
4. **Research and Development:** Robotics companies and agricultural researchers use fruit plucking robots for experimental purposes. These robots help in developing and testing new harvesting technologies, fine-tuning fruit detection algorithms, and optimizing various aspects of fruit harvesting.
5. **Home Gardens:** Although not as common, some smaller-scale robotic solutions have been developed for home gardeners. These robots can assist in harvesting fruits in residential settings, making gardening more accessible and efficient.
6. **Commercial Food Processing:** Ripe fruit plucking robots can be integrated into larger food processing facilities to streamline the process of collecting fruits for further processing, sorting,

or packaging. This can be particularly useful for fruit juice production, canning, and other food industries.

7. Large-Scale Agricultural Operations: On extensive agricultural properties with multiple crops and specialized fruit varieties, robots can provide scalability and adaptability. They can handle various types of fruits and adapt to different orchard layouts.

8. High-Value and Fragile Fruits: Ripe fruit plucking robots are especially valuable for high-value fruits that require gentle handling, such as berries, figs, and cherries. These robots are designed to minimize fruit damage and maximize the quality of the harvest.

9. Reducing Harvesting Costs For labor-intensive crops with high labor costs, robots offer a solution to reduce the expenses associated with human labor. This is particularly important in regions facing labor shortages or high labor costs.

10. Quality Control and Data Collection Beyond harvesting, fruit plucking robots often come equipped with sensors and cameras that can assess the quality of fruits. This data can be used for yield estimation, quality control, and farm management.

11. Fruit plucking robots are widely used in large-scale orchards and fruit farms, where they efficiently harvest a variety of fruits, reducing labor costs and ensuring that ripe fruits are collected at the right time.

12. In vineyards, these robots are invaluable for harvesting grapes, which require precise timing for wine production. They minimize damage to vines and enhance the quality of grapes picked.

13. Fruit plucking robots find applications in greenhouses, where they harvest fruits like tomatoes and cucumbers. Their ability to work in confined spaces makes them ideal for controlled environments.

14. Robotics companies and agricultural researchers utilize fruit plucking robots for experimentation and technological development, allowing for the testing and optimization of various harvesting techniques.

15.While less common, smaller-scale fruit plucking robots are designed for home gardeners, making fruit harvesting in residential settings more efficient and accessible, benefiting hobbyist gardeners.

Fruit plucking robots have the potential to transform fruit harvesting in agriculture, addressing a range of challenges while improving efficiency and fruit quality. As technology continues to advance, their applications are likely to expand to more agricultural and horticultural settings.

1.5 Limitations

While ripe fruit plucking robots offer numerous benefits, they also have certain limitations and challenges that need to be addressed for optimal performance. Here are some common limitations:

1. **Initial Cost:** The upfront cost of purchasing and deploying fruit plucking robots can be significant. This initial investment may be a barrier for smaller farms or agricultural businesses with limited budgets.
2. **Technical Complexity:** The complexity of the technology can be a limitation, as it may require specialized training and maintenance expertise. Farmers and operators need to have a certain level of technical proficiency to manage and troubleshoot these robots effectively.
3. **Adaptability:** While modern robots are becoming more versatile, they may not be as adaptable as human labor in certain situations. Robots may struggle to handle unexpected challenges or unique orchard layouts, which require manual intervention or adjustments.
4. **Fruit Detection Accuracy:** The accuracy of fruit detection algorithms is crucial for efficient harvesting. Suboptimal algorithms can result in missing ripe fruits or the picking of unripe ones. Enhancing the accuracy of these systems is an ongoing challenge.
5. **Handling Fragile Fruits:** Some fruit plucking robots may still struggle with handling extremely fragile fruits, like soft berries, without causing damage. Further advancements in gripper technology and manipulation are needed to handle these fruits with care.

6. **Energy Efficiency:** Energy consumption is a concern, as robots rely on batteries or power sources for operation. Extending the operating duration and improving energy efficiency is a continuous challenge.

7. **Obstacle Detection and Navigation:** While robots are equipped with obstacle detection technology, there is room for improvement in navigating complex orchards. This includes safely avoiding unexpected obstacles and dealing with uneven terrain.

8. **Harvesting Speed:** The speed at which fruit plucking robots operate may not match the pace of skilled human laborers. Speed enhancements are essential to make robot-assisted harvesting as time-efficient as manual harvesting.

9. **Fruit Sorting:** In some cases, robots may struggle with accurate sorting of fruits based on quality criteria. Improving the sorting capability of robots to meet industry standards is an ongoing focus.

10. **Maintenance and Repairs:** Like any machinery, fruit plucking robots require maintenance and occasional repairs. Keeping them in top working condition can be a logistical challenge, particularly for larger fleets of robots.

11. **Scale and Cost-Effectiveness:** For small or medium-sized farms, the cost-effectiveness of deploying robots may not be as evident compared to large-scale commercial operations. Smaller farms may not have the resources or fruit quantities to justify the investment.

12. **Regulatory and Ethical Concerns:** The introduction of robots into agriculture may raise concerns about potential job displacement and ethical issues related to replacing human labor. Addressing these concerns and regulations surrounding autonomous agricultural robots is an ongoing challenge.

Despite these limitations, ongoing research and development in the field of agricultural robotics are actively addressing many of these challenges, making ripe fruit plucking robots increasingly viable and effective solutions for modern agriculture.

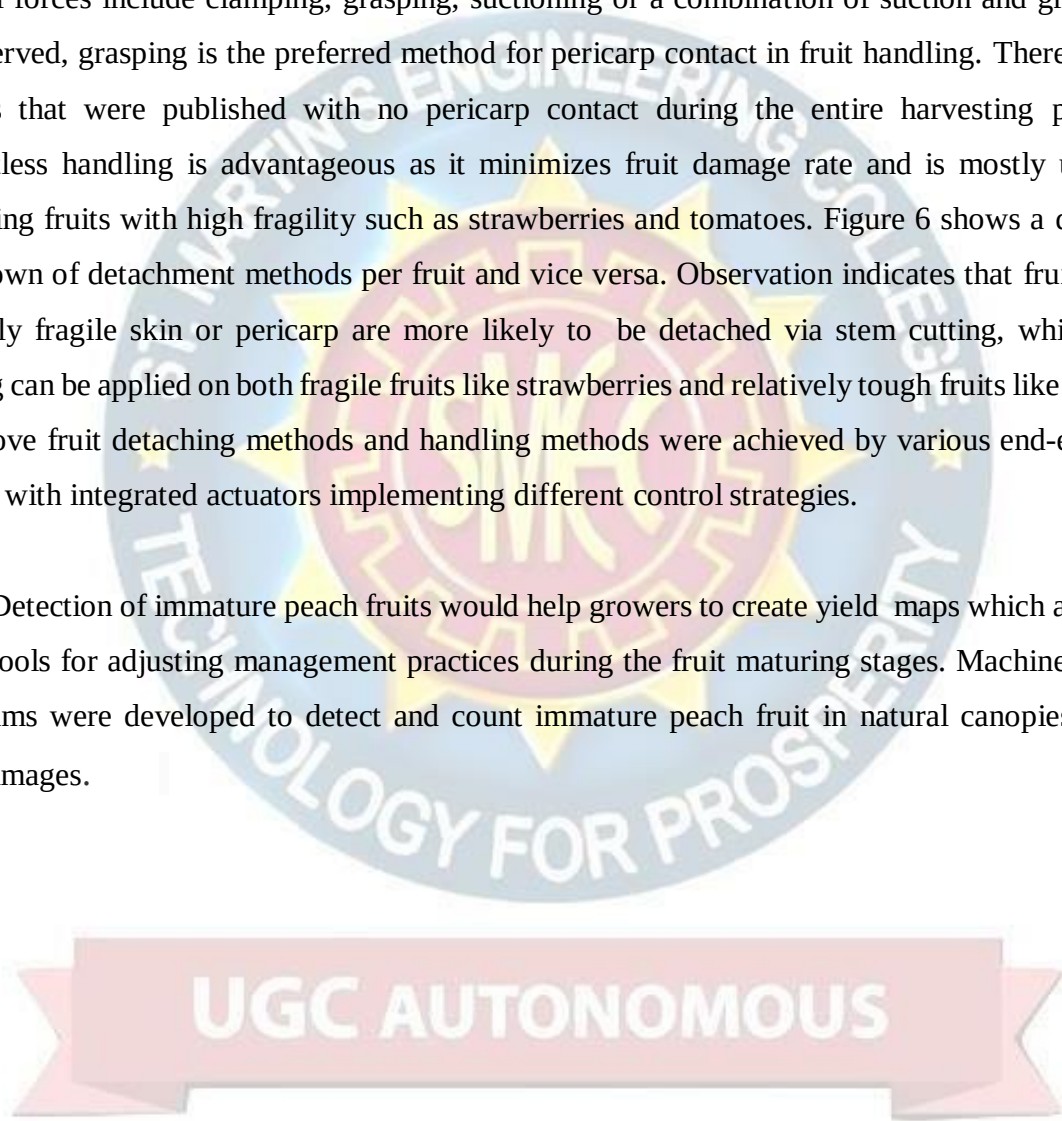
1.6 Summary

The occlusion of fruit from leaves stems, and other obstacles presents a challenge to harvesting robots in grasping and path planning. To simplify the problem, many published works require modification or simplification of the environment, such as applying proper supports to the plant for better lighting as well as better accessibility and visibility for robotic systems. Only 8 of the applications claimed that their robot is applied in unmodified environments. For soft-obstacle occlusions such as leaves and stems, several approaches to temporarily move them aside have been demonstrated, such as using mechanical means to gather up leaves so that the target fruit can be fully exposed to the camera or utilizing dual-arm cooperation to imitate human movements of pushing leaves to the side. For non-compliant obstacles such as branches and trellis, or complex structures in general, a multiple Degrees of Freedom manipulator with a properly designed path planning algorithm is usually implemented to avoid obstacles. The number of Doff implemented amongst researchers varies, including six, seven and nine. However, it is observed that a high Doff manipulator tends to increase the system complexity and thus may lead to lower picking efficiency. For example, Kang's 6 DOF manipulator reported a 6.5 s cycle time, while both Abundant Robotics' and Robotics' low Doff straight-line approach manipulator achieved less than 2 s cycle time. This approach resorts to using vision algorithms to filter out difficult-to-reach fruits, and those not filtered are assumed to be graspable using the straight-line path. Such methods have been applied to harvesting tomatoes, apples, and oranges. The final mode of occlusion via fruit clustering is a significant challenge for harvesting robots in which no reliable solution to the problem has been found to date. Many researchers have tried various methods to separate the clustering of ripe and unripe fruits, which is commonly seen in strawberry harvesting. Methods such as applying compressed air to blow adjacent fruits away, or the use of active obstacle-separation algorithms to isolate fruits presented only a 50–75% harvest success rate. This rate worsens to just 5% if the target fruit is completely occluded by unripe fruit.

Create a suction force to extract the fruit without direct contact with either the stem or fruit. shows the stem cut and fruit twist tend to be two major detaching methods, with corresponding average harvest success rates presented in where applications applied fruit twisting and stem cut recorded 75% and 65% average harvest rate respectively. Among the four systems adopted where stem pull is the detachment method, only two systems reported a 90% average

harvest success rate. However, because the data sample is small, it is premature to claim that any detachment method is superior to another, especially due to the sparsity of the fruit damage data that is currently provided in the literature. Fruit handling methods were also analyzed, with 27 of the 47 applications applying various forms of external force on the fruit. These are shown in where external forces include clamping, grasping, suctioning or a combination of suction and grasping. As observed, grasping is the preferred method for pericarp contact in fruit handling. There are 20 systems that were published with no pericarp contact during the entire harvesting process. Contactless handling is advantageous as it minimizes fruit damage rate and is mostly used in harvesting fruits with high fragility such as strawberries and tomatoes. Figure 6 shows a detailed breakdown of detachment methods per fruit and vice versa. Observation indicates that fruits with relatively fragile skin or pericarp are more likely to be detached via stem cutting, while fruit twisting can be applied on both fragile fruits like strawberries and relatively tough fruits like apples. The above fruit detaching methods and handling methods were achieved by various end-effector designs with integrated actuators implementing different control strategies.

Detection of immature peach fruits would help growers to create yield maps which are very useful tools for adjusting management practices during the fruit maturing stages. Machine vision algorithms were developed to detect and count immature peach fruit in natural canopies using colour images.



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CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY:

s.no	Author	Title	Year	contribution
1.	A. S. Teimouri	A review of vision-based fruit detection for robotic harvesting applications	2020	Offered a comprehensive review of vision-based fruit detection systems, summarizing their use in various robotic harvesting applications.
2.	N. I. Zarifneshat	A survey of fruit harvesting robots: Recent trends and challenges	2020	Conducted a comprehensive survey of fruit harvesting robots, summarizing recent trends and discussing the challenges in the field.
3.	Y. Gao	Robotic systems for automatic strawberry harvesting: Challenges and developments	2019	Focused on the challenges and developments in robotic systems for automatic strawberry harvesting, a particularly delicate task
4.	D. Lee	Developing an autonomous apple harvesting robot by combining low-cost, high-accuracy GPS and IMU sensors	2019	Described the development of an autonomous apple harvesting robot, highlighting the use of GPS and IMU sensors for navigation and fruit detection
5.	M. H. S. Jois and M. A. Hossain	Vision-based robotic fruit detection and localization for apple harvesting	2018	Developed a vision-based system using RGB-D cameras for apple detection and localization

Table no 2.1

6.	S. M. Sabzi and M. T. Hossain	Design and development of a novel robotic end-effector for grape-picking using compliant mechanisms	2018	Designed a novel robotic end-effector using compliant mechanisms for gentle grape picking, reducing fruit damage
7.	M. Jamzad	Automatic apple harvest: Fruit detachment and catching system	2017	Developed a system for both fruit detachment and catching, optimizing the apple harvesting process
8.	G. Alemdar and A. G. Yilmaz	Computer vision-based multiple fruit detection for robotic harvesting	2017	Proposed a computer vision system for detecting and localizing multiple fruits, addressing the challenge of variable fruit distribution
9.	J. A. Claver	Robotics in horticulture: A review	2017	Provided an overview of robotics applications in horticulture, including fruit harvesting, and discussed challenges and future prospects
10.	G. V. Khot	A review on advances in automated weed detection and its impact on farming	2017	Examined the role of automated systems in agriculture, including fruit plucking robots, in improving farming efficiency
11.	J. Fang	A vision-based ripe fruit detection system for apple harvesting robots	2016	Designed a vision-based system specifically for detecting ripe apples and guiding harvesting robots
12.	P. Gomez	Lidar-based fruit detection for robotic apple harvesting	2016	Demonstrated the use of Lidar technology for fruit detection, with a focus on apple harvesting using robotic systems
13.	A. Alenya	Grape's vision-based detection and localization for an autonomous robotic system	2015	Developed a vision-based system for detecting and localizing grapes, which is crucial for automated grape harvesting

Table no 2.2

2.2 SURVEY CONCLUSION

The landscape of ripped fruit plucking robots, designed to automate the complex and labour-intensive process of fruit harvesting, has witnessed substantial progress and innovation in recent years. Through an in-depth examination of existing research and development efforts, several prominent trends and conclusions come to the forefront. One of the standout themes is the extensive reliance on vision-based systems for fruit detection and location. These systems leverage advanced imaging techniques and sophisticated algorithms to discern ripe fruits with a high degree of accuracy. The use of computer vision plays a pivotal role in enabling these robots to differentiate between ripe and unripe fruits, a crucial aspect of efficient and selective harvesting. Another critical aspect of this field revolves around the design and development of grippers and end-effectors. Achieving a balance between secure fruit gripping and minimal damage during the plucking process is of paramount importance. Researchers have explored a wide array of gripper designs, including soft, flexible grippers, suction-based systems, and more traditional mechanical grippers, all with the overarching goal of ensuring that fruit damage is kept to a minimum during the harvesting process. Moreover, the control and navigation systems are central to the success of these robotic harvesters. Autonomous navigation and precise control systems are key components that underpin the operation of these robots. Advanced control algorithms, encompassing methodologies like fuzzy logic and machine learning, are incorporated to facilitate exact positioning and manipulation of the robot's end-effector, ultimately ensuring the efficient and accurate harvesting of ripe fruits. The literature survey further highlights a trend of specialization, with many studies focusing on specific crops, such as sweet peppers, apples, strawberries, and citrus fruits. This specialization is vital, as different crops possess unique characteristics that necessitate tailored solutions for effective harvesting. This tailored approach acknowledges that each fruit type presents its own set of challenges and requires dedicated research and engineering efforts.

Despite the substantial progress made in this field, it is essential to acknowledge that numerous challenges remain. These include the need for enhanced adaptability to the ever-changing environmental conditions of agricultural settings, the pursuit of faster detection and harvesting speeds to boost overall efficiency, and the imperative to drive down costs for practical adoption on farms.

CHAPTER 3

METHODOLOGY

Developing a ripe fruit-plucking robot is a multifaceted endeavor that demands a structured approach. The methodology for creating such a robot begins by clearly defining the project's scope and objectives. This initial stage involves setting specific parameters, including the types of fruits to be harvested, their ripeness criteria, and the design of the plucking mechanism. Following this, market research and requirements analysis are essential to understand user needs and existing solutions, providing a foundation for design decisions. With a clear vision in place, the conceptual design phase focuses on outlining the primary components, such as sensors, actuators, and control systems. This step is crucial in establishing the robot's core architecture. Subsequently, the mechanical design stage involves detailed drawings and specifications for the robot's physical structure, considering factors like fruit varieties and sizes. The heart of the robot lies in its sensors and vision system, which must be adept at fruit detection and ripeness assessment. Machine learning algorithms are often employed to recognize fruit based on factors like color, size, firmness, and scent. The ripeness detection capabilities are crucial for selective harvesting. Actuation and control are then addressed, with careful consideration given to the choice of actuators and the development of precise control systems. These systems enable the robot to carry out plucking actions accurately and efficiently. Software development is a vital parallel process, focusing on creating the control software for the robot, including algorithms for motion control, path planning, and communication with the ripeness detection system.

Ensuring the safety of both the robot and its surroundings is paramount. This entails the incorporation of safety measures like emergency stop mechanisms, collision detection, and obstacle avoidance systems. As the design progresses, prototyping is essential. Building a physical prototype allows for practical testing and validation of the plucking mechanism and ripeness detection algorithms. This iterative process continues, with extensive testing and adjustments made based on real-world performance. User interface and communication components are designed to provide an accessible and user-friendly means of controlling and monitoring the robot. Additionally, integrating the robot with existing agricultural automation systems is crucial if it's intended for large-scale fruit harvesting. Field testing takes the robot out of the lab and into real orchards. These trials evaluate the robot's performance, efficiency, and durability under practical conditions. Valuable data and user feedback are collected and analyzed to identify areas for improvement.

FLOW CHART

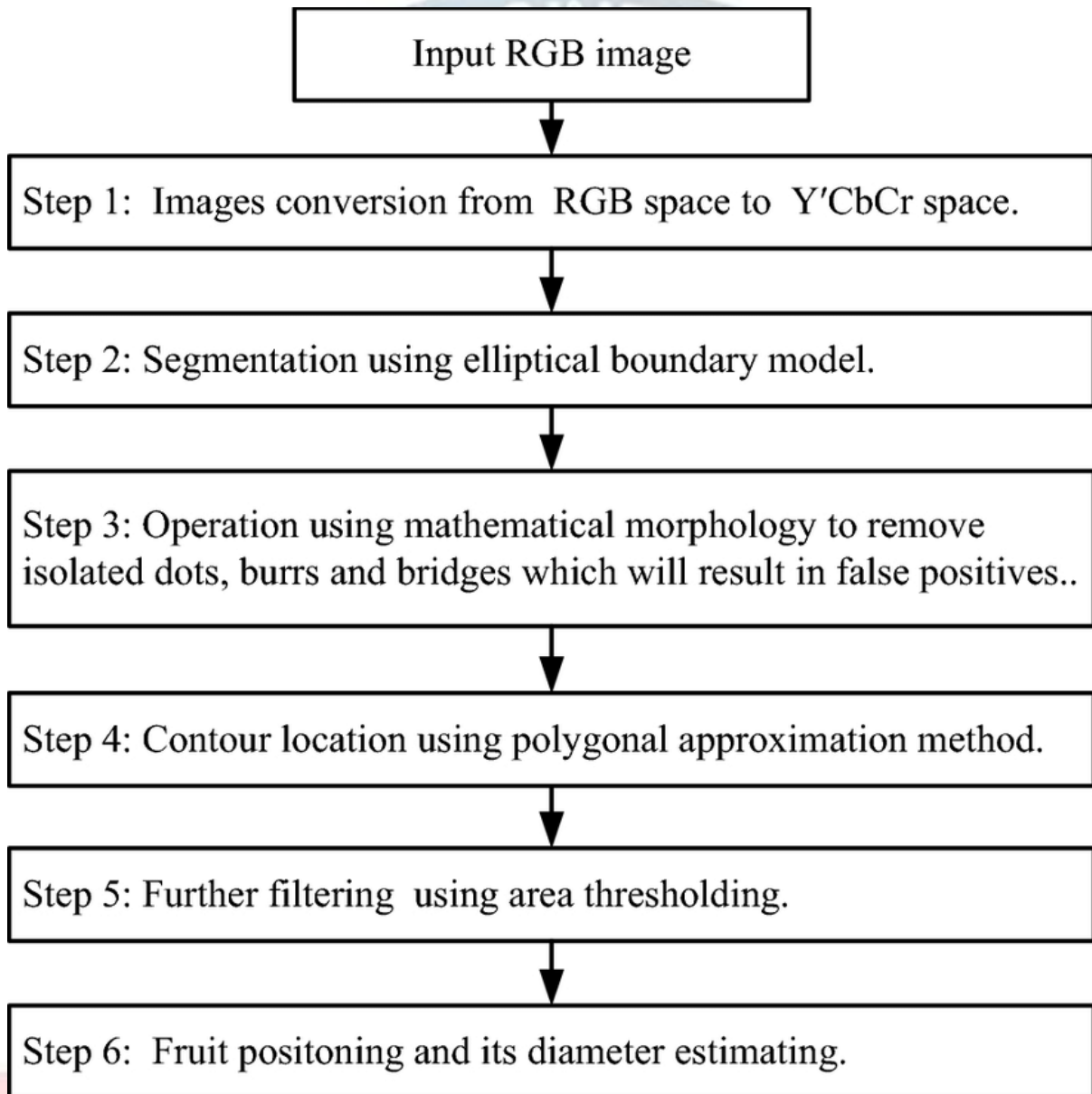


Fig 3.1 Flowchart

harmoniously with other technologies on the farm, enhancing its overall efficiency. Field testing is critical for assessing real-world performance, enabling the collection of data on plucking efficiency, reliability, and fruit quality. Continuous data collection and analysis inform further improvements and optimizations, making the robot more effective and reliable. Documentation, training, and scaling up production are essential for commercial deployment, while compliance with regulatory standards and ongoing maintenance and support provide a framework for long-term use. Lastly, a commitment to continuous improvement ensures that the robot adapts to evolving needs and technological advancements. Developing a ripe fruit plucking robot is a complex and multifaceted undertaking that necessitates a systematic and methodical approach. The process commences with a clear definition of the project's scope and objectives, meticulously detailing the types of fruits to be harvested, the specific criteria for ripeness detection, and the intricacies of the plucking mechanism. This initial phase serves as the compass, steering the project in the right direction and forming the basis upon which all subsequent design and development will be predicated. Extensive market research and rigorous requirements analysis are then undertaken, providing a comprehensive understanding of the existing solutions, user needs, and the constraints within which the robot must operate. The insights garnered from this research play an indispensable role in shaping critical decisions at every stage of the design process. Following this, the conceptual design phase takes center stage, encompassing the establishment of the robot's fundamental architecture, defining key components, such as sensors, actuators, and control systems, and mapping out the high-level design decisions.

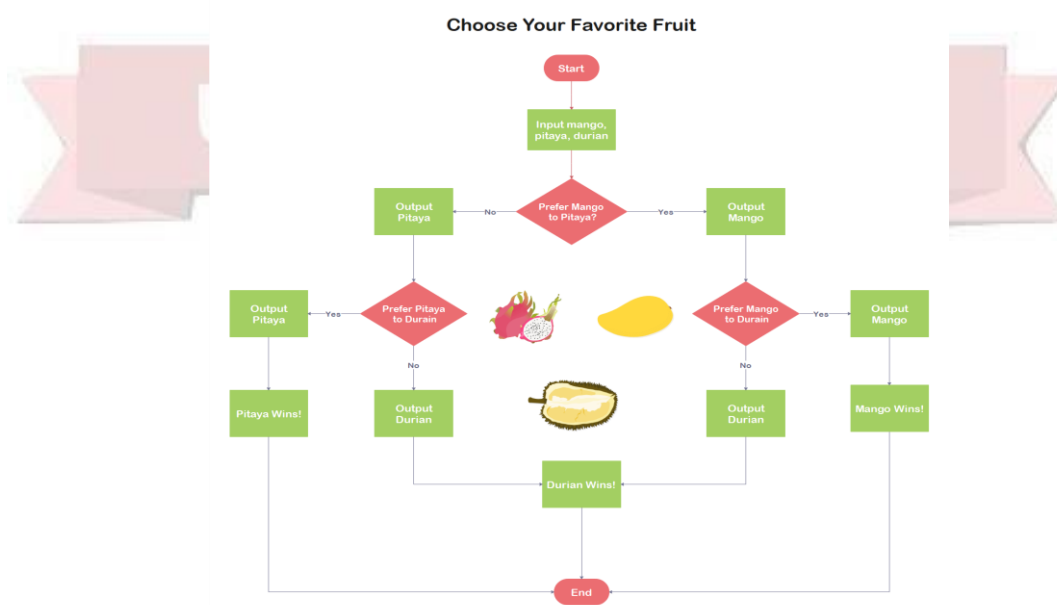


Fig 3.2 Flowchart

Subsequently, the mechanical design phase delves into the intricate specifics, yielding detailed drawings and specifications for the robot's physical structure. This phase is where considerations for the robot's versatility in accommodating various fruit types and sizes are meticulously addressed, ensuring adaptability to diverse orchard environments. At the core of the robot's functionality lies its sensory and vision system, where sophisticated machine learning algorithms come into play. These algorithms enable the robot to identify fruits based on a gamut of factors, including color, size, firmness, and scent, thereby providing the robot with the necessary acumen to discriminate ripe from unripe fruit with precision. Actuation and control systems are meticulously chosen and developed to ensure the robot's precision and accuracy in performing its tasks. This involves selecting the appropriate actuators, designing control algorithms to govern precise and synchronized motion, and developing path planning algorithms to facilitate efficient navigation through orchards. The software development phase follows suit, encompassing the programming of the robot for motion control, interpretation of high-level commands, and decision-making based on the ripeness data gleaned from the vision and sensor systems. Ensuring the safety of the robot and its surroundings is imperative. Emergency stop systems, collision detection mechanisms, and obstacle avoidance strategies are methodically integrated to preserve the robot's integrity and prevent any inadvertent damage. Once the design elements are in place, the prototype is constructed. Testing rigorously ensues, aimed at evaluating the effectiveness of the plucking mechanism, reliability of the control software, and the robustness of the robot's design. These tests not only validate the design but also provide crucial insights for refinement and further development.



FIG: 3.1 Parts of Robot

User interfaces are designed with user-friendliness in mind, facilitating easy control and monitoring of the robot's operations, including options for remote operation and seamless data exchange and communication. The integration of the robot with existing agricultural automation systems is paramount, ensuring that it harmoniously interacts with other technologies on the farm, ultimately enhancing its overall efficiency. To truly gauge real-world performance, field testing becomes an integral part of the development process. It provides an authentic environment for assessing the robot's plucking efficiency, reliability, and the quality of harvested fruit. Continuous data collection and rigorous analysis not only inform further refinements but also drive ongoing improvements and optimizations, making the robot increasingly effective and reliable. Additionally, comprehensive documentation, user training, and plans for scaling up production are pivotal for eventual commercial deployment. Regulatory compliance with established standards and a robust system for ongoing maintenance and support form the bedrock for the robot's long-term utilization.

A perpetual commitment to continuous improvement guarantees that the robot remains adaptable to evolving needs and capitalizes on emerging technological advancements, ensuring its sustained relevance and efficacy. Developing a ripe fruit plucking robot demands a meticulous and structured approach that unfolds through several interrelated phases. Commencing with a crystal-clear definition of the project's scope and objectives, it is imperative to specify the targeted fruit types, the criteria for ripeness determination, and the design intricacies of the plucking mechanism. This foundational step serves as the North Star, guiding all subsequent efforts, and laying the groundwork upon which the robot's design, development, and performance will be built. In-depth market research and exhaustive requirements analysis follow, offering invaluable insights into the landscape of existing solutions, comprehending user needs, and understanding the operational boundaries and constraints the robot will encounter. The fruits of this research act as a cornerstone for making informed decisions at every juncture of the design process. The conceptual design phase takes center stage, giving birth to the high-level architectural blueprint of the robot. It is during this phase that the essential components are defined, including sensors, actuators, and control systems, and where the grand design decisions are charted. The decisions made in this phase set the overarching structure within which the robot's physical form and functionality will evolve. After the conceptual design, the mechanical design phase ventures into detailed drawings and specifications,

dictating the physical manifestation of the robot. This phase is pivotal for addressing the robot's adaptability to a multitude of fruit types and sizes, ensuring versatility and suitability to diverse orchard environments. The robot's cognitive ability to discern ripe from unripe fruit is situated at the heart of its functionality. Leveraging intricate machine learning algorithms, it becomes proficient in identifying fruits based on a multitude of characteristics, encompassing color, size, firmness, and fragrance. This cognitive capacity bestows the robot with the acumen necessary for precise and selective plucking.

Actuation and control systems come under meticulous scrutiny next. The apt choice of actuators and the development of precise control algorithms are critical to ensuring the robot's movements are accurate, dependable, and coordinated. The robot's path planning algorithms are also engineered to navigate through orchards with efficiency, avoiding obstacles and hazards. Simultaneously, software development assumes a central role in the robot's functionality. Motion control software is meticulously crafted, enabling the robot to interpret high-level commands and translate them into precise motor commands.



CHAPTER 4

SYSTEM ANALYSIS AND DESIGN

4.1 EXISTING SYSTEM

As of my last knowledge update in September 2021, the existing systems of ripe fruit plucking robots demonstrated a range of sophisticated features and technologies, although there was no standardized system due to the varying designs and functionalities developed by different companies and organizations. These robots typically incorporated advanced sensing technology, such as cameras for visual fruit detection based on color and shape, and often employed LiDAR or ultrasonic sensors for obstacle detection and orchard mapping, which allowed them to autonomously navigate the orchard. Control systems included onboard computers with complex algorithms for fruit recognition, path planning, and real-time decision-making. Many of these robots also featured wireless communication capabilities, enabling remote monitoring and control by operators. Gripper mechanisms, a crucial aspect of these robots, varied based on the type of fruit to be harvested. Soft and compliant grippers designed to handle delicate fruits without causing damage were common, as were vibration or shaking mechanisms to gently detach fruits from branches. Power sources ranged from traditional batteries to more sustainable options like rechargeable batteries and even solar panels, contributing to prolonged operational durations in the field. Data collection was another essential element, with sensors assessing fruit quality, recording yield data, and creating orchard maps. This information played a vital role in yield estimation and orchard management. Safety features, including obstacle-detection sensors, were crucial to avoid collisions and ensure safe operation. Some systems even offered customization options, such as interchangeable grippers and settings that could be adjusted to accommodate different fruit varieties and orchard layouts. Additionally, a few existing systems featured fruit sorting and handling mechanisms, minimizing the need for additional manual labor during post-harvest processing. Machine learning and AI capabilities were integrated to improve performance over time, allowing for better fruit recognition and decision-making. It's important to acknowledge that the field of fruit plucking robots is dynamic, and newer systems likely have emerged with additional features and improvements, all aimed at addressing the labor-intensive nature of fruit harvesting while maintaining fruit quality and contributing to greater efficiency in agriculture. The field of ripe fruit plucking robots has continued to advance since my last update in September 2021. These robots have become even more sophisticated and versatile in their capabilities. Modern

based on color and shape but can also assess fruit quality with increasing accuracy. LiDAR sensors and GPS technology have been further refined, enabling precise navigation and mapping of orchards, even in complex layouts.

In addition, the control systems of these robots have become more intelligent and adaptive, incorporating machine learning and artificial intelligence algorithms that continuously improve fruit recognition and decision-making. This has led to enhanced accuracy in detecting ripe fruits and optimal harvesting times, thereby minimizing waste, and increasing overall efficiency. Furthermore, gripper mechanisms have evolved to handle a broader range of fruit types, sizes, and shapes, with softer, more compliant materials that minimize fruit damage. Some robots have introduced advanced vibration and shaking mechanisms that apply gentle forces to trees or branches, prompting ripe fruits to fall into collection bins with minimal bruising. Energy efficiency has been a key focus, with many robots now incorporating advanced battery technology and, in some cases, renewable energy sources such as solar panels for recharging. This not only reduces operational costs but also aligns with sustainability objectives. The concept of human-robot collaboration has gained prominence, where robots work alongside human labor to optimize the harvesting process. This approach addresses the potential job displacement concerns and leverages the strengths of both humans and robots in a complementary manner. Overall, ripe fruit plucking robots have witnessed significant technological advancements, making them more efficient, adaptable to various fruit types and orchard conditions, and environmentally friendly. These robots are poised to play an increasingly vital role in modern agriculture, alleviating labor shortages and contributing to improved efficiency and sustainability in fruit harvesting practices.

4.2 PROPOSED SYSTEM

While my last knowledge update was in September 2021, I can provide a concept of a proposed system for ripe fruit plucking robots based on the trends and developments in the field up to that point. The proposed system for ripe fruit plucking robots aims to incorporate cutting-edge technologies and address several of the limitations of existing systems. In this advanced system, the primary focus would be on further enhancing the accuracy and efficiency of fruit detection and harvesting. This would involve the integration of even more advanced computer vision algorithms that not only recognize ripe fruits based on color and shape but can also assess the quality of fruits in real-time, identifying imperfections and ensuring that only the highest-quality produce is picked. Navigation and obstacle avoidance systems would be further refined to enable seamless operation in complex orchard layouts and adverse weather conditions.

These robots would have the capability to adapt to changing terrains and adjust their movements in real-time, ensuring safe and efficient navigation.

The proposed system would also place a strong emphasis on energy efficiency and sustainability. It would incorporate the latest developments in battery technology, allowing for extended operational durations and quicker recharging. Additionally, the integration of renewable energy sources, such as advanced solar panels, would make these robots eco-friendlier and reduce their carbon footprint. To address the challenge of handling fragile fruits, the gripper mechanisms would be designed with even greater precision and delicacy. They would be adaptable to a wider range of fruit types and sizes, ensuring that fruits are plucked without damage or bruising.

Machine learning and artificial intelligence would continue to be a cornerstone of the proposed system. These technologies would enable the robots to continually improve their fruit recognition capabilities and decision-making processes. Over time, the robots would become more autonomous and adaptive, learning from their experiences, and optimizing their performance. Human-robot collaboration would be another integral aspect of the proposed system. The robots would work in tandem with human labor, augmenting the workforce and ensuring a harmonious coexistence. Human operators would oversee and fine-tune the robots' actions, intervening when necessary and addressing any unforeseen challenges. Overall, the proposed system for ripe fruit plucking robots envisions a future where technology and agriculture harmoniously coexist. These robots would not only revolutionize fruit harvesting in terms of efficiency and yield but also contribute to sustainable agricultural practices, addressing labor shortages and minimizing waste. The system represents an exciting step forward in the evolution of agricultural robotics, promising a brighter and more productive future for fruit cultivation. The proposed system for ripe fruit plucking robots represents a leap forward in the fusion of cutting-edge technology and agricultural efficiency. In this vision of the future, these robots are equipped with advanced computer vision algorithms, enabling them to identify ripe fruits with unparalleled accuracy. Beyond color and shape, these algorithms assess the quality of each fruit in real-time, ensuring that only the freshest and most pristine produce is selected for harvesting.

Navigation and obstacle avoidance systems in the proposed system are refined to provide robots with a heightened level of adaptability. They can seamlessly navigate through complex orchards, efficiently adjusting their paths to accommodate varying terrains and weather conditions. This not only enhances their efficiency but also ensures their operation in diverse agricultural settings. Energy efficiency is a core tenet of the proposed system, with robots benefiting from state-of-the-

art battery technology, leading to extended operational periods and reduced downtime for recharging. Furthermore, sustainable energy sources like advanced solar panels are integrated to power these robots, making them more environmentally friendly and cost-effective. The delicate handling of fruits is a paramount concern in the proposed system, with gripper mechanisms engineered for precision and gentleness. These mechanisms adapt to a broad spectrum of fruit varieties and sizes, promising that the harvesting process remains gentle and damage-free. Machine learning and artificial intelligence continue to drive innovation in the proposed system. These robots continuously evolve, learning from their experiences to optimize fruit recognition and decision-making. As they accumulate data, they become increasingly autonomous, responding adeptly to evolving orchard conditions. Human-robot collaboration is a defining feature, aligning with the ethos of combining the strengths of both. These robots function harmoniously with human labor, augmenting the workforce and ensuring a seamless, efficient operation. The role of human operators transforms into one of supervision and fine-tuning, allowing for intervention when needed and overcoming unforeseen challenges.

In sum, the proposed system for ripe fruit plucking robots epitomizes the future of precision agriculture. It promises efficiency, sustainability, and the highest quality of produce, all while addressing labor shortages and reducing waste. This vision underscores the pivotal role of technology in shaping the agriculture of tomorrow, offering a promising and productive path for the cultivation of fruits

4.3 DEVELOPMENT OF RIPPED FRUIT PLUCKING ROBOT



Fig 4.1 Ripped Fruit Plucking Robot

The design of a ripe fruit plucking robot is a multifaceted endeavor that encompasses various critical components and considerations. At its core, the mechanical design is a pivotal aspect, which often features a flexible robotic arm equipped with an adaptable end effector, such as a gripper or specialized plucking tool. This mechanical structure must facilitate precise and controlled movements to reach fruits at varying heights and angles. An integral part of the robot's design is its sensory and vision system, which typically comprises cameras, including high-resolution and potentially 3D cameras, to provide vision capabilities for locating and identifying ripe fruits accurately. Furthermore, the inclusion of force sensors in the gripper or plucking tool is instrumental in determining the optimal plucking force, ensuring that fruits are detached without causing damage to either the fruit or the plant. The heart of the robot's functionality lies in its ability to detect fruit ripeness. This necessitates the development of sophisticated algorithms, leveraging sensor data on color, size, firmness, and scent. Machine learning models are commonly employed to recognize fruits and assess their ripeness accurately. Meanwhile, actuation and control systems play a vital role in executing precise and coordinated movements. The selection of suitable actuators, such as servos or motors, is pivotal for ensuring controlled motion of the robot's joints and end effector. In tandem with this, control algorithms are developed to manage the robot's motion, enabling it to navigate orchards or fields with efficiency. Software development also encompasses decision-making processes based on ripeness data and communication between the robot's various components.

Safety remains paramount throughout the design process, with the integration of safety features like emergency stop buttons, collision detection, and obstacle avoidance mechanisms to prevent accidents and protect both the robot and its operators. Power supply considerations are equally vital, whether through battery systems or tethered power supplies, ensuring that the robot has sufficient energy to carry out extended operations in diverse outdoor conditions. User interfaces are designed to be user-friendly, allowing operators to control and monitor the robot easily. This interface may take the form of a graphical user interface (GUI) or a mobile app for remote operation. Furthermore, the robot's integration with existing automation systems is critical, fostering compatibility with farm management software and data analytics platforms. Durability plays a crucial role in the robot's design, with the selection of weather-resistant materials to withstand various environmental conditions. The robot should be capable of functioning in a range of weather conditions and terrains, offering robustness in its outdoor operations. The design process involves prototyping and rigorous testing in controlled

environments to validate the robot's reliability and performance. Field testing is subsequently conducted in real fruit orchards to assess the robot's efficiency under practical, real-world conditions, accounting for uneven terrain and varying fruit densities.

4.4 DEVELOPMENT OF VARIOUS COMPONENTS



Fig: 4.4 Upper Arm And Gripper

Robotic arms, often simply referred to as robot arms, are fundamental components of robotic systems renowned for their versatility and precision. These mechanical systems are engineered to replicate the functionalities of a human arm, enabling them to perform a wide array of tasks, from lifting and transporting objects to executing intricate and precise movements. Central to their design are multiple joints that provide degrees of freedom, allowing the arms to move in various directions. Common configurations include arms with three, four, five, or six degrees of freedom,

each affording a specific range of motion. Typically, robotic arms are composed of individual links, which represent the arm's segments, connected by joints, the structural components that facilitate movement. Actuators, often in the form of motors, hydraulic systems, or other driving mechanisms, empower the joints and drive the arm's movement. At the arm's terminus, an end effector, or tool, is often mounted to perform specialized tasks, such as gripping, welding, or surgical procedures. These versatile machines find application in a wide spectrum of industries, including manufacturing, healthcare, agriculture, and space exploration. With varying sensor systems and programming methods, robotic arms can perform tasks autonomously or with human guidance. Collaborative robots, or cobots, equipped with safety features and sensors, can work safely alongside humans. Ongoing advancements in materials, control systems, and machine learning continue to enhance the capabilities of robotic arms, making them indispensable in modern automation and industry, contributing to increased efficiency and productivity across a myriad of fields.



Fig: 4.5 cutter

The chassis of a ripe fruit plucking robot constitutes the fundamental framework upon which the entire robot is built. It serves as the structural backbone, housing and supporting the essential components and systems that make up the robot. When designing the chassis for such a robot, several critical considerations come into play. The design must account for the robot's intended operational environment in fruit orchards, characterized by uneven terrain and exposure to various environmental conditions. Therefore, sturdiness and durability are paramount, ensuring the chassis can withstand potential physical stresses and endure the rigors of outdoor use. Material selection is a crucial aspect, with lightweight yet robust options such as aluminum or carbon fiber being common choices, offering the ideal combination of strength and weight for

efficient operation. Customizability is another key feature of the chassis, allowing for the precise integration and secure housing of the robot's components, including the power source, sensors, actuators, control systems, and the plucking mechanism. Weight distribution is carefully considered to maintain the robot's stability and prevent tipping, particularly during operations that involve carrying harvested fruit. Modularity, a hallmark of versatile robotic systems, is essential in chassis design, as it simplifies maintenance and facilitates component upgrades without necessitating extensive redesign. The chassis should promote maneuverability, enabling the robot to efficiently navigate orchards, reach fruit-bearing trees, and travel between them.

Sufficient ground clearance is essential to prevent the robot from getting stuck on uneven terrain, tree roots, or debris found in orchards. Furthermore, the chassis must provide protection to internal components, safeguarding them against the elements by incorporating weatherproofing and dust-resistant features. Safety features, such as collision detection and avoidance systems, may be integrated into the chassis to safeguard the robot from obstacles in the orchard and prevent potential accidents. Scalability is an additional consideration, with the chassis design allowing for adjustments in size or configuration to meet varying operational needs or accommodate different fruit tree types and sizes. In summary, the chassis is the cornerstone of a ripe fruit plucking robot's design, supporting 31 protecting the technology that drives its operation while ensuring it can navigate orchards with precision and resilience, ultimately enhancing fruit harvesting efficiency and productivity.

4.5 WORKING FLOW CHART

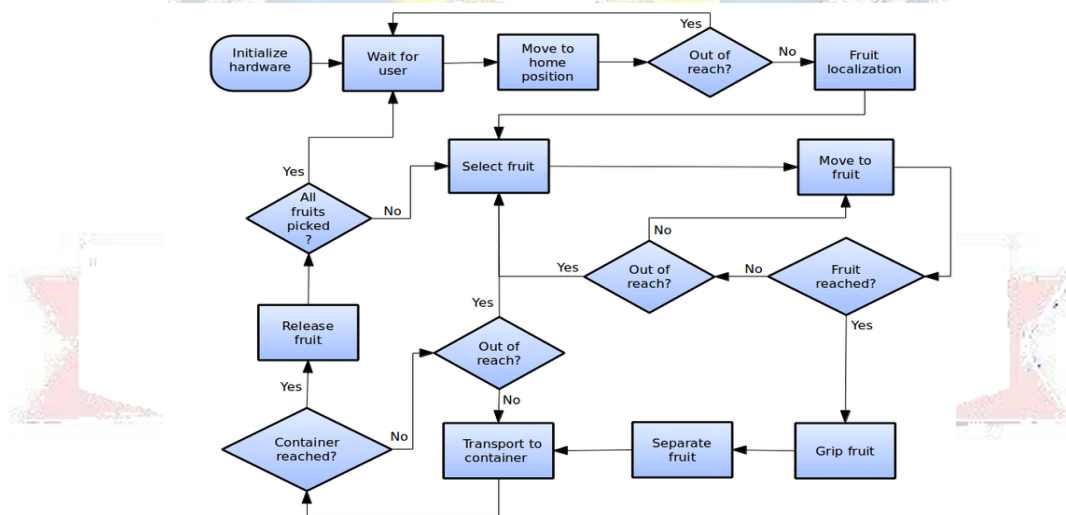


Fig 4.6 Flow Chart

1.Start: The flowchart begins with the starting point, representing the initiation of the process with Power Supply

2. Power to Arduino UNO: This step involves power supply to Arduino UNO board, where all major connection of robot is established.

Arduino is considered as the brain of the Biped Robot and it is responsible for controlling various part of Robot including servo motor, the program is written in the processor of this arduino.



Fig 4.5.1 Arduino Board

3. Raspberry pi

The Raspberry Pi is a low cost, credit-card sized computer that plugs into a computer monitor or TV, and uses a standard keyboard and mouse. It is a capable little device that enables people of all ages to explore computing, and to learn how to program in languages like Scratch and Python. It's capable of doing everything you'd expect a desktop computer to do, from browsing the internet and playing high-definition video, to making spreadsheets, word- processing, and playing games.

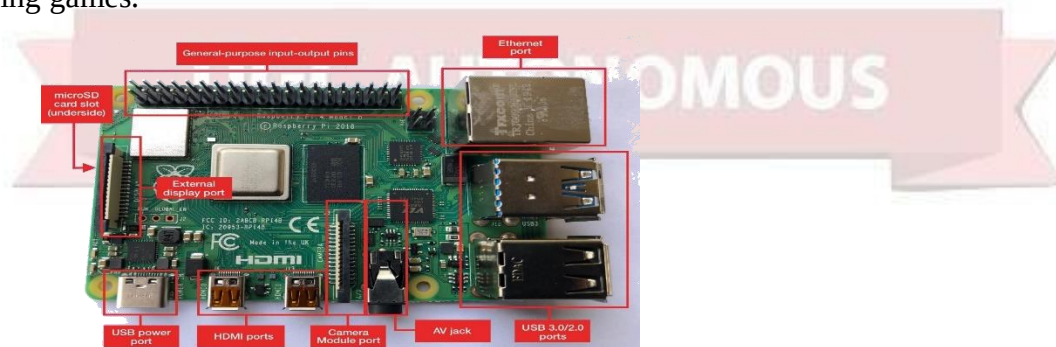


Fig 4.5.2 Raspberry pi

4. Signal to Servo Motor Board: The Arduino after getting command from user translates the electric signals to servo board where all servo motors are attached with help of pin. Each servo motor has unique pin and therefore command is given to a particular pin to move servo motor.



Fig 4.5.3 Signal Board

5. Servo Motors: The electrical energy coming by the circuit junction to the desired Servo motor and then the motor transforms the electric current into mechanical motion and gives the desired motion given by the User



Fig 4.5.4 Servo Motor

6. ACTUATOR: An actuator is a component of a machine that produces force, torque, or displacement, usually in a controlled way, when an electrical, pneumatic, or hydraulic input is supplied to it in a system (called an actuating system). An actuator converts such an input signal into the required form of mechanical energy. It is a type of transducer.



Fig 4.5.5 Actuator

4.6 Source code

4.5.1 Code for robot arm movement:

```
#define BLYNK_TEMPLATE_ID "TMPL3n6dr569c"
#define BLYNK_TEMPLATE_NAME "Arm"
#define BLYNK_AUTH_TOKEN "pALM-e9bFPiEepYFkPYieqi03Q_JrDZq"
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <Servo.h>

// You should get Auth Token in the Blynk App.
// Go to the Project Settings (nut icon).
char auth[] = "pALM-e9bFPiEepYFkPYieqi03Q_JrDZq";
// Motor Pins
#define MOTOR1A D0
#define MOTOR1B D1
#define MOTOR2A D2
#define MOTOR2B D3
#define MOTOR3A D4
#define MOTOR3B D8
#define MOTOR4A A0
#define MOTOR4B D9 // RX pin
#define PWMM D10 // TX pin
// Servo Pins
#define SERVO1_PIN D5
#define SERVO2_PIN D6
#define SERVO3_PIN D7
Servo servo1;
Servo servo2;
Servo servo3;
void setup()
{
  // Debug console
  Serial.begin(9600);
  pinMode(D10,OUTPUT);
  analogWrite(D10, 100);
```

```

Blynk.begin(auth, "Jishnu", "1234567890");
pinMode(MOTOR1A, OUTPUT);
pinMode(MOTOR1B, OUTPUT);
pinMode(MOTOR2A, OUTPUT);
pinMode(MOTOR2B, OUTPUT);
pinMode(MOTOR3A, OUTPUT);
pinMode(MOTOR3B, OUTPUT);
pinMode(MOTOR4A, OUTPUT);
pinMode(MOTOR4B, OUTPUT);

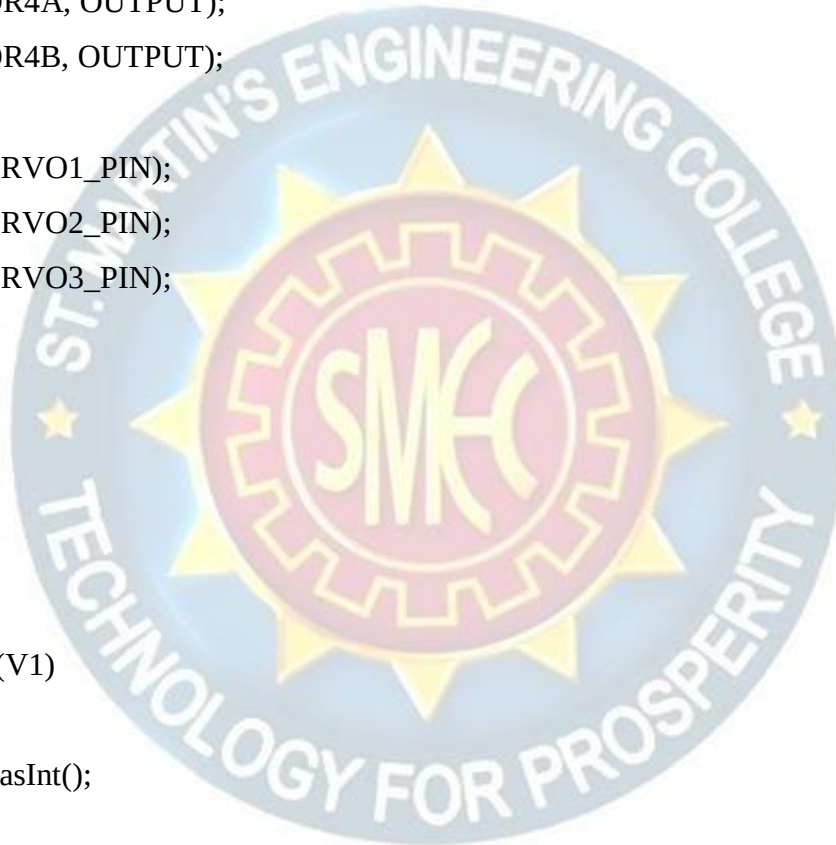
servo1.attach(SERVO1_PIN);
servo2.attach(SERVO2_PIN);
servo3.attach(SERVO3_PIN);
}
void loop()
{
  Blynk.run();
}

BLYNK_WRITE(V1)
{
  int y = param[0].asInt();

  // Forward and Backward Movement
  if (y > 200) {
    Serial.println("Moving Forward");
    // Move Forward
    digitalWrite(MOTOR3A, HIGH);
    digitalWrite(MOTOR3B, LOW);
    digitalWrite(MOTOR4A, LOW);
    digitalWrite(MOTOR4B, LOW);

    digitalWrite(MOTOR4A, HIGH);
    digitalWrite(MOTOR4B, LOW);
  }
}

```



UGC AUTONOMOUS

```

}
else if (y < 100) {
  Serial.println("Moving Backward");
  // Move Backward
  digitalWrite(MOTOR3A, LOW);
  digitalWrite(MOTOR3B, HIGH);
  digitalWrite(MOTOR4A, LOW);
  digitalWrite(MOTOR4B, HIGH);
}
else {
  Serial.println("Stop");
  // Stop
  digitalWrite(MOTOR3A, LOW);
  digitalWrite(MOTOR3B, LOW);
  digitalWrite(MOTOR4A, LOW);
  digitalWrite(MOTOR4B, LOW);
}
}

```

```

BLYNK_WRITE(V2)
{
  int x = param[0].asInt();

```

```

  // Left and Right Movement

```

```

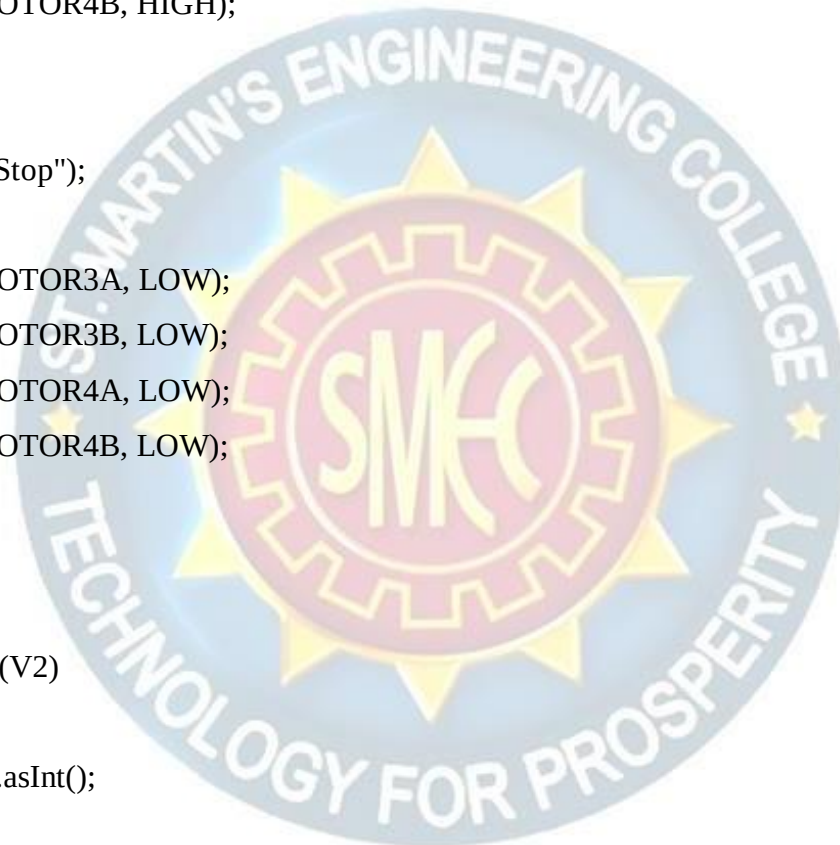
  if (x > 200) {
    Serial.println("Moving Right");
    // Move Right
    digitalWrite(MOTOR3A, HIGH);
    digitalWrite(MOTOR3B, LOW);
    digitalWrite(MOTOR4A, LOW);
    digitalWrite(MOTOR4B, HIGH);
  }

```

```

  else if (x < 100) {
    Serial.println("Moving Left");

```



UGC AUTONOMOUS

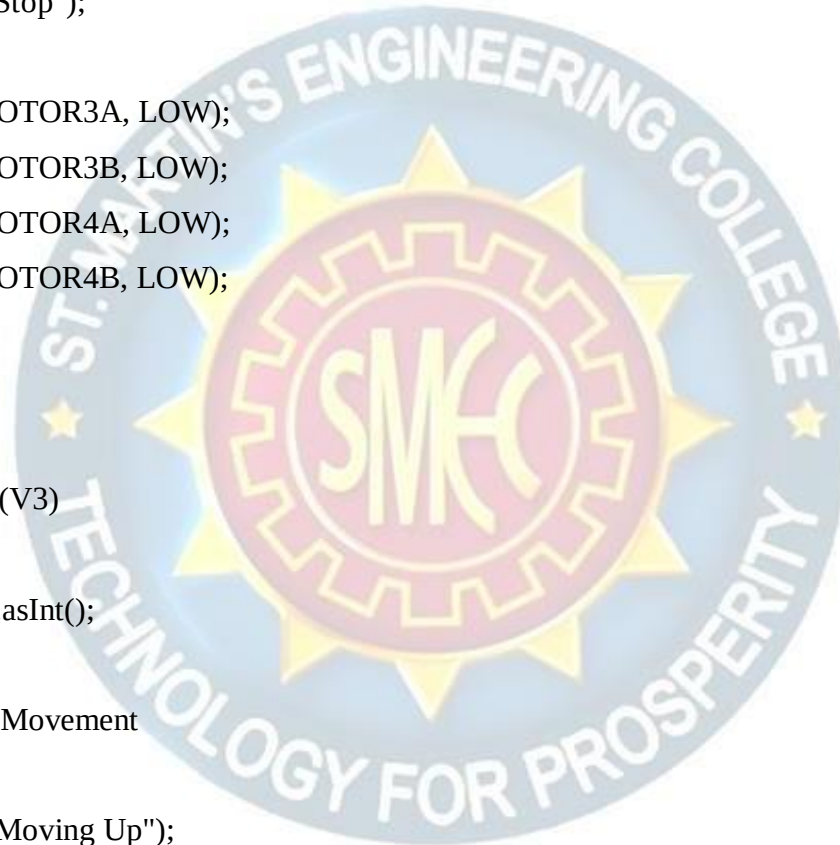

```

// Move Left
digitalWrite(MOTOR3A, LOW);
digitalWrite(MOTOR3B, HIGH);
digitalWrite(MOTOR4A, HIGH);
digitalWrite(MOTOR4B, LOW);
}
else {
  Serial.println("Stop");
  // Stop
  digitalWrite(MOTOR3A, LOW);
  digitalWrite(MOTOR3B, LOW);
  digitalWrite(MOTOR4A, LOW);
  digitalWrite(MOTOR4B, LOW);
}
}

BLYNK_WRITE(V3)
{
  int z = param[0].asInt();

  // Up and Down Movement
  if (z > 200) {
    Serial.println("Moving Up");
    // Move Up
    digitalWrite(MOTOR1A, HIGH);
    digitalWrite(MOTOR1B, LOW);
  }
  else if (z < 100) {
    Serial.println("Moving Down");
    // Move Down
    digitalWrite(MOTOR1A, LOW);
    digitalWrite(MOTOR1B, HIGH);
  }
  else {

```



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```

Serial.println("Stop");
// Stop
digitalWrite(MOTOR1A, LOW);
digitalWrite(MOTOR1B, LOW);
}
}

```

```

BLYNK_WRITE(V4)
{
  int z = param[0].asInt();

  // Up and Down Movement
  if (z > 200) {
    Serial.println("Moving Up");
    // Move Up
    digitalWrite(MOTOR2A, HIGH);
    digitalWrite(MOTOR2B, LOW);
  }
  else if (z < 100) {
    Serial.println("Moving Down");
    // Move Down
    digitalWrite(MOTOR2A, LOW);
    digitalWrite(MOTOR2B, HIGH);
  }
  else {
    Serial.println("Stop");
    // Stop
    digitalWrite(MOTOR2A, LOW);
    digitalWrite(MOTOR2B, LOW);
  }
}
}

```

```

BLYNK_WRITE(V5)
{
  int angle1 = param.asInt();

```

```

servo1.write(angle1);
Serial.print("Servo 1 Angle: ");
Serial.println(angle1);
}
BLYNK_WRITE(V6)
{
  int angle2 = param.asInt();
  servo2.write(angle2);
  Serial.print("Servo 2 Angle: ");
  Serial.println(angle2);
}

BLYNK_WRITE(V7)
{
  int angle3 = param.asInt();
  servo3.write(angle3);
  Serial.print("Servo 3 Angle: ");
  Serial.println(angle3);
}

```

Code description:

Blynk Configuration:

The code begins with the configuration of Blynk, an IoT platform that allows for remote control and monitoring of connected devices. It includes the template ID and authentication token required for connecting the device to the Blynk project.

Header Files:

The necessary header files, including ESP8266WiFi.h, BlynkSimpleEsp8266.h, and Servo.h, are included to enable Wi-Fi communication, Blynk functionality, and servo motor control, respectively.

Pin Definitions:

Pins are defined for motor control (MOTOR1A, MOTOR1B, etc.) and servo motors (SERVO1_PIN, SERVO2_PIN, SERVO3_PIN). These pins correspond to the physical connections between the microcontroller (likely an ESP8266-based board) and the motors/servos.

Setup Function:

The setup() function initializes serial communication for debugging purposes and configures the pins for motor and servo control. It also establishes a connection to the Blynk server using the provided authentication credentials.

Loop Function:

The loop() function is responsible for continuously running the Blynk service, which maintains the connection to the Blynk server and handles incoming commands from the Blynk app.

Blynk Write Functions:

Several BLYNK_WRITE() functions are defined to handle incoming commands from the Blynk app. Each function corresponds to a specific virtual pin (V1-V7) defined in the Blynk project. These functions interpret the received values and control the motors and servos accordingly to achieve desired arm movements and servo positions.

For example, BLYNK_WRITE(V1) handles commands for forward and backward movement, BLYNK_WRITE(V2) handles commands for left and right movement, and so on.

The digitalWrite() and analogWrite() functions are used to set the motor direction and servo positions based on the received commands.

Servo Control:

Additional BLYNK_WRITE() functions (V5-V7) are included specifically for controlling the angles of the servo motors. These functions receive angle values from the Blynk app and set the corresponding servo positions accordingly using the servo.write() function.

5.7.2 Code for Object Detection

```
from cvlib.object_detection import YOLO
import cv2
cap=cv2.VideoCapture(0)
weights="yolov4-tiny-custom_3000.weights"
config="yolov4-tiny-custom.cfg"
labels="obj.names"
count=0
while True:
    ret,img=cap.read()
    count += 1
    if count % 2 != 0:
        continue
    img=cv2.resize(img,(680,460))
    yolo = YOLO(weights, config,labels)
```

```

bbox, label, conf = yolo.detect_objects(img)
fruits=yolo.draw_bbox(img, bbox, label, conf, write_conf=True)
cv2.imshow("fruits",img)
if cv2.waitKey(1)&0xFF==27:
    break
elif cv2.getWindowProperty('fruits', cv2.WND_PROP_VISIBLE) < 1:
    break

```

Code description:

This Python code snippet utilizes the cvlib library to perform object detection using the YOLO (You Only Look Once) algorithm on a live video feed from a webcam. Below is a breakdown of the code:

Importing Libraries:

from cvlib.object_detection import YOLO: Imports the YOLO object detection model from the cvlib library.

import cv2: Imports the OpenCV library for computer vision tasks.

Setting up Video Capture:

cap = cv2.VideoCapture(0): Initializes video capture from the default camera (index=0).

Defining YOLO Configuration:

weights = "yolov4-tiny-custom_3000.weights": Path to the pre-trained YOLO weights file.

config = "yolov4-tiny-custom.cfg": Path to the YOLO configuration file.

labels = "obj.names": Path to the file containing class labels.

Object Detection Loop:

while True:: Initiates an infinite loop for continuous video processing.

ret, img = cap.read(): Captures a frame from the video feed.

count += 1: Increments the frame counter.

if count % 2 != 0:: Skips every other frame to reduce processing load.

img = cv2.resize(img, (680, 460)): Resizes the frame to a standard size for YOLO processing.

yolo = YOLO(weights, config, labels): Initializes the YOLO object detector with provided weights, configuration, and labels.

bbox, label, conf = yolo.detect_objects(img): Detects objects in the resized frame and returns bounding box coordinates, labels, and confidence scores.

fruits = yolo.draw_bbox(img, bbox, label, conf, write_conf=True): Draws bounding boxes around detected objects on the frame.

cv2.imshow("fruits", img): Displays the processed frame with bounding boxes.

if cv2.waitKey(1) & 0xFF == 27:: Breaks the loop if the 'Esc' key is pressed.

elif cv2.getWindowProperty('fruits', cv2.WND_PROP_VISIBLE) < 1::

Breaks the loop if the display window is closed.

This code continuously captures frames from the webcam, performs object detection using YOLO, draws bounding boxes around detected objects, and displays the processed frames in real-time. It provides a simple implementation for real-time object detection using YOLO in Python with OpenCV and cvlib.



Fig 4.6.1 view of Ripped fruit

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CHAPTER 5 SYSTEM REQUIREMENTS AND SPECIFICATIONS

5.1 BASIC BLOCK DIAGRAM :

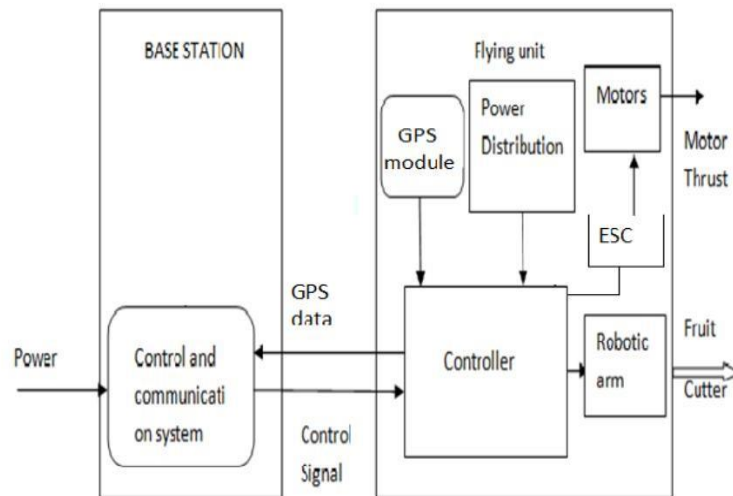


Fig 5.1 Block Diagram

5.2 COMPONENTS OF RIPPED FRUIT PLUCKING ROBOT:

- Robotic Arm
- End-Effector/Gripper
- Sensors
- Computer Vision System
- Control System
- Power Source
- Communication Module
- Artificial Intelligence (AI) Algorithms
- Mechanical Structure
- User Interface
- Safety Features
- Localization and Navigation System

GRIPPER

Soft grippers are a type of robotic gripper designed to handle objects gently and adapt to various shapes and sizes. They use flexible and deformable materials, such as elastomers, to grasp objects, making them well-suited for applications where fragile or irregularly shaped items need to be manipulated. Here are some key characteristics and features of soft grippers:



Fig: 5.2.1 Gripper

ROBOTIC ARM

A robotic arm is a versatile mechanical system designed to replicate the functions of a human arm with precision and dexterity. Comprising a series of interconnected links and joints, robotic arms can execute a wide range of tasks, from simple pick-and-place actions to complex, delicate maneuvers. The number of degrees of freedom, typically ranging from three to six, determines the arm's flexibility and range of motion. These arms are commonly equipped with actuators, such as motors or servos, which drive the joints, allowing for controlled movement. The end effector, often a tool or gripper, is attached to the arm's terminus and is customizable for specific tasks, making robotic arms adaptable for various applications. Robotic arms find use in diverse industries, including manufacturing, where they perform tasks like welding, assembling products, and handling materials on production lines. In healthcare, robotic arms are used in surgical procedures, enabling precision and minimally invasive interventions. Additionally, they play a vital role in fields like aerospace, where they aid in assembly and maintenance, and in agriculture, particularly in the development of fruit plucking robots for harvesting operations. As technology continues to advance, robotic arms are becoming increasingly sophisticated, with the integration of advanced sensors and machine learning algorithms, enabling them to carry out more complex and delicate tasks with greater accuracy and efficiency. Their versatility and adaptability make them indispensable tools for automation and industry, contributing to increased

productivity and precision in a wide range of applications.



Fig: 5.2.2 Robotic Arm

CHASIS

The chassis of a fruit plucking robot represents the sturdy and essential structural framework that underpins the robot's functionality in the complex task of fruit harvesting. This crucial component provides the necessary support for mounting and housing all the intricate systems and components that enable the robot to operate efficiently in orchards and fruit-laden environments. Designing the chassis of a fruit plucking robot demands careful consideration of several key factors. Given the often uneven and challenging terrains of orchards, the chassis must be engineered for durability, strength, and resilience to withstand the rigors of outdoor use. Material selection plays a pivotal role, with lightweight but robust options such as aluminum or carbon fiber being favored choices, offering the necessary strength without excessive weight. Customizability is integral, as the chassis needs to accommodate and securely house diverse components, including power sources, sensors, actuators, control systems, and the plucking mechanism. Weight distribution is a critical aspect to ensure the robot's stability and prevent tipping while carrying harvested fruit.

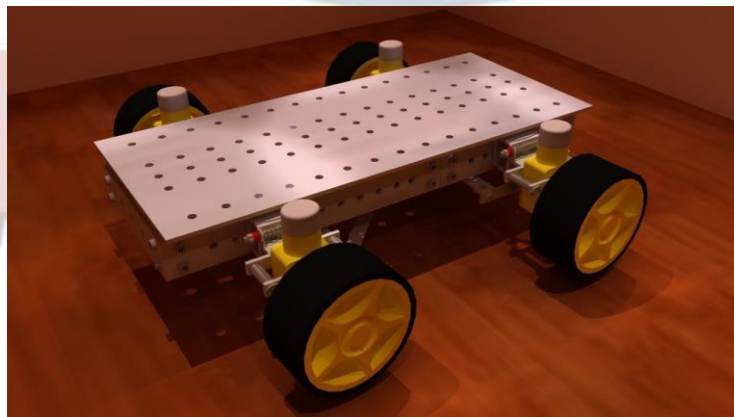


Fig 5.2.3 Chassis

Sensor

To detect and localize the fruit, the images of the environment which record the information of targets and backgrounds are recorded and analyzed, utilizing either two- dimensional or three- dimensional imaging sensors. 2D imaging sensors record the two- dimensional information of the scene, typically in red, green, and blue colors, infrared, spectral sensors, or a combination of any of them. RGB cameras detect the target fruits from the background by extracting and analyzing the fruit properties like color, geometric shape, and texture by traditional machine learning methods or by using deep learning, which is known to be robust, accurate and efficient. However, RGB cameras are highly sensitive to illumination and detecting the target fruits that have similar color with background is also challenging. Spectral sensors that provide spectral and spatial information can be utilized to perform fruit detection based on the various reflectance at different wavelengths. Spectral imaging processing can potentially address the issue caused by the less distinct color signature of target fruits and background, but the time-consuming data collection and image processing limits its practical application. Thermal sensors record the temperature information of the objects and canopies, which can benefit the detection and segmentation of fruits from a less distinct background. This is because the fruits can absorb more heat and emit more heat compared to the canopies. However, its limitations are apparent when detecting fruits in shadowed areas located in the deep tree canopies. While 2D imaging sensors are low cost and easy to access, normally they alone cannot provide the accurate 3D spatial location of the target fruits in the work environment for robots utilizing the selective harvesting method. Several types of 3D visual sensors exist on the market, such as stereo cameras, Light Detection and Ranging sensors and RGB-D cameras. The practical applications of these cameras are shown in Fig. 3, with the maximum number of applications focused on the stereo cameras. Stereo cameras take images using two or multiple RGB cameras separated by a fixed distance. The images are then fused, with depth calculated through triangulation. While effective, this sensor is slow due to the imaging fusing process and requires frequent calibration, which limits its real-time practicality. LiDAR is a remote range sensor that captures spatial information using pulsed laser reflection to generate a two or three dimensional point cloud. These sensors are robust and generally perform well under strong natural light, but they cannot provide color information that is critical for vision-based fruit detection. Although LiDAR can be combined with RGB cameras, data fusion can be slow, and the high price of millimeter accurate LiDAR limits its wider application. RGB-D cameras are functionally identical to 2D RGB cameras, but can also process depth images simultaneously,

offering a real-time stream of RGB images fused with depth information. These cameras use either time-of-flight or structured light methods to calculate the depth distance. Tofu sensors utilize an infrared light emitter, which emits pulses of light to measure the distance to the target. In the structure light technique, infrared light is emitted in a pattern, and depth information can be calculated based on the level of distortion of this light pattern when returning to the receiver. Compared with stereo cameras, RGB-D cameras are far superior in localization accuracy, robustness, and computational efficiency, and are much cheaper than LiDAR sensors for the same accuracy, albeit with a smaller viewing range. These advantages of the RGB-D camera over the others make it more suitable for practical applications and it has been an essential component of vision systems in agriculture applications. However, the RGB-D camera will experience performance degradation at a close scanning distance or under extreme weather conditions like strong sunlight. This is because of the lower quality of depth images generated under the above-mentioned scenarios.



Fig: 5.2.4 Sensor

Computer vision system

A Computer Vision System is a technology that enables computers to interpret and understand visual information from the world, much like humans do with their eyes. It involves the use of cameras, image sensors, and advanced algorithms to process, analyze, and make sense of visual data. These systems start by acquiring visual data through cameras or imaging devices, which can be static images or dynamic video sequences. The acquired data often undergoes preprocessing steps, such as noise reduction and image enhancement, to improve its quality for analysis. Subsequently, features like edges, corners, textures, and colors are extracted from the images to represent specific patterns or characteristics. Computer Vision Systems can detect and identify objects or regions of interest within images or videos, making them essential for tasks

like face detection or object recognition. Additionally, they can track the movement of objects or subjects over time, which is particularly valuable in surveillance and robotics. These systems also excel in image classification, where they categorize images into predefined classes or labels, as well as image segmentation, which divides an image into meaningful regions based on various criteria like color, intensity, or texture. Pattern recognition is another vital function, enabling these systems to recognize shapes or patterns in images, a capability crucial for medical diagnosis and scene understanding.

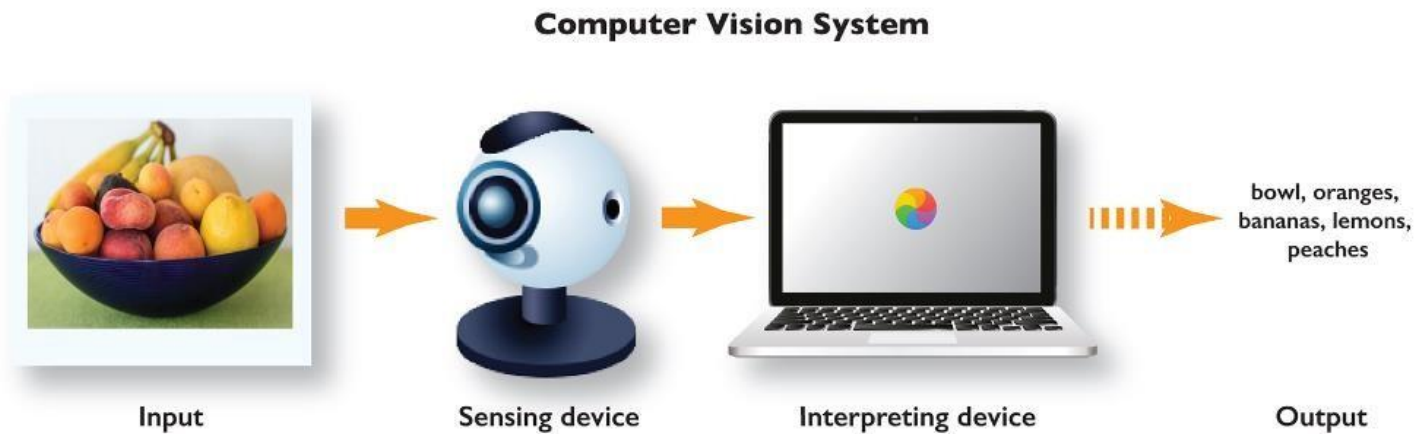


Fig: 5.2.5 Computer Vision System

Power source / battery:

Lead-acid batteries are a well-established and widely used type of rechargeable battery known for their reliability and robustness. These batteries are composed of lead dioxide as the positive plate and sponge lead as the negative plate, immersed in a solution of diluted sulfuric acid, serving as the electrolyte. Lead-acid batteries are commonly employed in a multitude of applications, including automotive vehicles, uninterruptible power supplies (UPS), and renewable energy storage systems. One of the notable features of lead-acid batteries is their ability to deliver a high current output, making them suitable for applications requiring a sudden burst of power, such as starting an engine in a car. They are also cost-effective and have a relatively long lifespan, especially in applications with regular charging and discharging cycles. However, lead-acid batteries have some limitations. They are relatively heavy and have a lower energy density compared to newer battery technologies like lithium-ion. Additionally, they require regular maintenance, including topping off the electrolyte and ensuring proper ventilation to manage hydrogen gas emissions during charging. Despite these limitations, lead-acid batteries remain essential in various industries due to their durability and affordability. As technology continues to advance, there is ongoing research and development to enhance their performance and address their environmental impact, making them a key player in the energy storage landscape,

particularly in applications where cost-effectiveness and reliability are paramount.



Fig: 5.2.6 Battery

Communications module:

A communication module, often referred to as a communication device or communication interface, is a crucial component in modern technology systems, enabling the exchange of data and information between different devices, systems, or networks. These modules are designed to establish and manage connections, ensuring that data can be transmitted, received, and processed effectively. Communication modules come in various forms, such as wired and wireless modules. Wired communication modules use physical connections like Ethernet cables or USB connections, offering reliable and high-speed data transfer in applications like computer networking. Wireless communication modules, on the other hand, utilize radio frequencies to transmit data without physical connections, making them ideal for mobile devices, IoT applications, and remote monitoring. These modules often incorporate specific communication protocols, such as Wi-Fi, Bluetooth, Zigbee, cellular networks, or proprietary protocols, to establish connections and enable data exchange. Communication modules are fundamental in a wide range of applications, from smartphones and laptops to IoT devices, industrial automation, and smart home systems, allowing devices to connect and interact with each other and with the broader internet. In summary, communication modules play a critical role in the connectivity of today's technology ecosystem, facilitating the seamless exchange of data and enabling the functioning of various interconnected devices and systems. They are instrumental in enabling modern applications and services to communicate and share information efficiently and in real-time.



Fig: 5.2.7 Communication Module

Control System:

A control system is an essential component in various engineering and industrial applications. It is designed to regulate and manage the behaviour of a system or process to achieve specific objectives. Control systems can be categorized into two main types: open-loop and closed-loop (or feedback) systems. In an open-loop control system, the output is not influenced by the system's actual performance. It relies on a predetermined input to produce an output, making it less adaptable to changes or disturbances in the system. These systems are often used for simple tasks where accuracy is not critical. Closed-loop control systems, on the other hand, incorporate feedback mechanisms to continuously monitor the system's performance. These systems adjust the control input based on the difference between the desired output and the actual output, ensuring that the system operates as intended. They are widely used in applications where precision and stability are crucial, such as industrial automation, aircraft navigation, and temperature regulation in heating and cooling systems.

Artificial Intelligence (AI) Algorithms:

Artificial Intelligence (AI) algorithms are the driving force behind the remarkable

capabilities of AI systems. These algorithms are sets of computational rules and procedures that enable machines to process data, learn from it, and make intelligent decisions or predictions. They underpin a wide range of AI applications, from machine learning to deep learning, natural language processing, and computer vision. Machine learning algorithms, for instance, allow systems to improve their performance over time by learning from data, while deep learning algorithms power advanced neural networks used in tasks like image and speech recognition. Natural language processing algorithms enable machines to understand and generate human language, revolutionizing chatbots, translation services, and sentiment analysis. Computer vision algorithms, on the other hand, enable the interpretation of visual data from images and videos, playing a critical role in tasks like facial recognition and object detection. These AI algorithms continue to evolve and find application in diverse fields, revolutionizing industries and transforming how we approach complex problem-solving and decision-making processes.



CHAPTER 6

CONCLUSION AND FUTURE SCOPE

CONCLUSION

In conclusion, the development of a ripe fruit plucking robot is a multifaceted and intricate process that demands a methodical and interdisciplinary approach. This robot is poised to revolutionize fruit harvesting in agricultural settings, offering precision, efficiency, and selectivity that human labor alone cannot achieve. The methodology presented here, from project inception to continuous improvement, underscores the complexity of designing such a robotic system. Throughout the development journey, the primary focus remains on addressing the specific needs of fruit growers, enhancing productivity, and ensuring the quality of harvested fruits. The robot's ability to distinguish ripe from unripe fruit through advanced sensory and vision systems is a game-changer, ensuring that only the most suitable fruits are plucked, minimizing waste, and optimizing crop yields. The integration of cutting-edge technologies, including machine learning, advanced sensors, and complex control systems, allows the robot to navigate orchards, adapt to variable environmental conditions, and collaborate seamlessly with existing agricultural automation systems. The emphasis on safety mechanisms and ongoing improvement underscores the commitment to operational excellence and the well-being of workers and property.

As the agriculture industry continues to evolve, the ripe fruit plucking robot represents a transformative leap forward, offering a solution to labor shortages and the need for efficient fruit harvesting. The potential impact of this technology is profound, leading to increased productivity, reduced food waste, and a more sustainable approach to fruit cultivation. The successful development of such robots signifies a milestone in the intersection of robotics and agriculture, promising a brighter and more efficient future for the fruit industry. The development of ripe fruit plucking robots represents a remarkable intersection of cutting-edge technology, agricultural innovation, and sustainability. These robots are poised to revolutionize the fruit harvesting industry, offering a host of benefits that extend far beyond mere efficiency. The methodology described is a comprehensive roadmap that guides the complex and multifaceted process of designing and implementing these robots. The significance of ripe fruit plucking robots cannot be overstated. These machines promise to alleviate some of the most pressing challenges facing the agricultural sector. From labor shortages and rising labor costs to

the need for increased production efficiency and reduced food waste, these robots stand as a beacon of hope. At their core, ripe fruit plucking robots are equipped with advanced sensory and vision systems, enabling them to discern ripe fruits from their unripe counterparts with remarkable precision. Machine learning algorithms play a pivotal role in this capability, as they continuously evolve and adapt, ensuring that the robot's performance remains at the cutting edge. The integration of sophisticated control systems, actuators, and meticulous mechanical design ensures that the robot can navigate orchards and pluck fruits with both speed and care. Furthermore, the safety features, including emergency stop systems and collision detection, offer peace of mind for both operators and fruit growers, as well as protection for the robot itself. These robots are not just mechanical marvels; they also exemplify the harmony between technology and agriculture. Their potential impact on the industry is vast, leading to increased productivity, reduced labor dependency, and more efficient resource use. The elimination of guesswork in fruit harvesting ensures that fewer fruits go to waste, contributing to sustainability and reducing environmental impact. As the ripe fruit plucking robots continue to evolve, they hold the promise of transforming the agricultural landscape. Their incorporation into existing automation systems offers not only efficiency but also scalability, adaptability, and the potential for data-driven decision-making in fruit production. In essence, these robots signify a significant milestone in the history of agriculture and automation. Their development and deployment serve as a testament to human ingenuity and innovation, addressing some of the most pressing challenges in the fruit industry while paving the way for a more sustainable, productive, and technology-driven future.

FUTURE SCOPE

The future scope of "ripped fruit plucking robots" holds significant promise in the field of agriculture and technology. Here are several key areas of potential development and impact:

1. Advanced Sensor Technology
2. Adaptability
3. Integration with Precision Agriculture
4. Environmental Sustainability
5. Data Analytics
6. Scalability
7. Global Adoption
8. Regulatory and Safety Considerations
9. Human-Machine Collaboration

CHAPTER 7

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